

Time Consistent Decisions with General Sets of Probabilities

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A broad literature in economics and computer science studies decisionmaking when an agent's belief is expressed as a set of probability measures. Under the condition of rectangularity, the natural approach of conditioning each probability measure on an observation leads to *time consistent* decisions: the optimal decision conditioned on the observation is also optimal a priori. This paper gives a general theory of time consistency beyond rectangularity and related conditions. We first demonstrate that if the objective is captured in a single real-valued operator, then the tower property of this operator is equivalent to time consistency. We then show an impossibility result beyond rectangularity: the tower property is generally incompatible with other natural axioms. As a result, there is no way to update a set of probability measures that leads to time-consistent decisions for any decision problem. This impossibility covers common proposals to move beyond rectangularity, such as maintaining weights for each probability measure. To resolve the impossibility, we give a novel time-consistent decision procedure for general sets of probability measures, based on the emerging literature of game-theoretic statistics. We conclude with several open directions.

1 Introduction

There is a rich literature on ways to capture a belief about a random future outcome Y . Aside from a single probability measure P , one of the most prominent approaches in economic theory, robust optimization, and finance, is to represent such a belief as a set of probability measures $\mathcal{P}$. Under this belief, the badness of a random future loss X , e.g. $X = L(Y)$, would be assessed via the upper expectation $\bar{\mathbb{E}}_{\mathcal{P}} X := \sup_{P \in \mathcal{P}} E_P X$. In economics and related fields, a common question to then ask is how to *update* such a belief $\mathcal{P}$ given new information Y_1 . Ideally such an update would lead to time-consistent decisions, meaning if a decision a is optimal when conditioned on every possible value of Y_1 , then a should be optimal before seeing Y_1 as well.

Several works explore specific suggestions for how update $\mathcal{P}$. Perhaps the most natural is *simple conditioning*, where one updates $\mathcal{P}$ to $\mathcal{P}|_{Y_1} := \{P(Y | Y_1)\}_{P \in \mathcal{P}}$. Here Grünwald and Halpern [12] show that under the condition of *rectangularity*, which roughly says that one can switch from any $P \in \mathcal{P}$ to any $Q \in \mathcal{P}$ upon seeing Y_1 , such an update does lead to time consistency. To address non-rectangular $\mathcal{P}$ where simple conditioning fails to be time consistent, works such as Halpern and Leung [13] consider a weighted version of $\mathcal{P}|_{Y_1}$, for example by tagging each distribution P with the likelihood $P(Y_1 = y_1)$ of the observed outcome.

While the existing literature explores specific suggestions for $\mathcal{P}[Y_1]$, and ways to operationalize them to make decisions, none lead to time consistency for all $\mathcal{P}$. Thus a fundamental general question remains open: is there a general way to update $\mathcal{P}$ that works for any $\mathcal{P}$? Specifically, can we define some object $\mathcal{P}[Y_1]$ for any set $\mathcal{P}$, and a method of assessing loss based on it, which leads to time-consistent decisions for every decision problem?

We answer this question negatively. In particular, we provide the following.

- (1) An axiomatic study of time consistency and equivalence to the tower property (§3). We first suppose that one wishes to make decisions from $\mathcal{P}[Y_1]$ using a conditional operator $\bar{\mathbb{E}}^c[X | Y_1]$, choosing the decision whose loss minimizes $\bar{\mathbb{E}}^c[\cdot | Y_1]$. Theorems 3.1 and 3.2 show that the tower property of $\bar{\mathbb{E}}^c$ is both necessary and sufficient for time consistency.
- (2) An impossibility result (§4). We give choices of $\mathcal{P}$ such that no conditional operator $\bar{\mathbb{E}}^c$ can satisfy the tower property, nor time consistency. In particular, there is no way to update $\mathcal{P}$ to some $\mathcal{P}[Y_1]$ so that one can make time-consistent decisions via an operator that depends only on $\mathcal{P}[Y_1]$ and $X | Y_1$. Beyond a counterexample, we give a broad class of $\mathcal{P}$ where this impossibility holds, including simple cases like sets of i.i.d. Bernoulli distributions (Theorem 4.6).
- (3) A novel time-consistent decision procedure for general $\mathcal{P}$ based on game-theoretic statistics (§5). From our impossibility result, we know that any time consistent operator $\bar{\mathbb{E}}^c[X | Y_1]$ must somehow depend on $\mathcal{P}$, X , and Y_1 more holistically. Using the theory of supermartingales, our approach is to add capital *accounts* for each $P \in \mathcal{P}$. Along with choosing actions in each round, the agent also chooses a gamble for each account P which is fair under P , making sure to have enough capital at the end to replicate the value of the loss for the chosen actions. We show that this account-valued operator satisfies a version of the tower property, and thus leads to time-consistent decisions.

Beyond a general impossibility, our results establish deep connections between time consistency, the tower property, and game-theoretic statistics. We conclude with several new avenues of inquiry enabled by these connections (§6).

1.1 Related literature

Our work relates to threads of research in four fields.

Computer science. This paper builds on Grünwald and Halpern [12], who study decision-making with sets of probabilities $\mathcal{P}$. They show that standard conditioning leads to *dilation*, a phenomenon where observing data makes the agent *less* certain and causes time inconsistency. While they characterize the conditions under which standard conditioning works, we ask whether time consistency can be recovered for *general* sets of probabilities by modifying the update procedure itself.

Motivated by these limitations, Halpern and Leung [13] propose *likelihood updating*: a process that maintains weights for each distribution in $\mathcal{P}$. Their main result characterizes the associated Minimax Weighted Expected Regret (MWER) decision rule. However, as we show, likelihood updating does not guarantee time consistency for general sets of probabilities (Theorem 4.6).

Economics. Time consistency is closely related to the economic concept of *dynamic consistency*. Epstein and Schneider [5] axiomatically characterize this property for Maxmin Expected Utility agents. They prove that dynamic consistency holds when the set of priors satisfies a structural condition, *rectangularity*. While this condition guarantees consistency under measure-by-measure Bayesian updating, it restricts the admissible beliefs.

When priors are not rectangular, Siniscalchi [20] proposes *consistent planning*, a backward induction approach that accepts dynamic inconsistency by choosing the best plan the future self will actually follow. In contrast, our work seeks to restore time consistency for general priors through an operator-based procedure rather than restricting beliefs or accepting inconsistent planning.

Finance. In mathematical finance, various notions of time consistency have been proposed for dynamic (time-dependent) financial risk measures, to capture the consistency of the induced risk preferences [2]. The tower property shows up frequently in these works; for example, Di Nunno and Øksendal [4] show that a version of time consistency is equivalent to the tower property, similar to our Theorem 3.2. Bartl [1] shows in turn that the tower property is equivalent to $\mathcal{P}$ satisfying rectangularity, where $\mathcal{P}$ is the set of probability measures in the dual (“robust”) representation of the risk measure. These works do not generally study time consistency outside of rectangularity, which given the form of financial risk measures (minimally satisfying Axioms 1 and 2 below), is consistent with our impossibility result in §4.

Game-theoretic statistics. Finally, as we discuss further in §6, our work relates in several ways to the emerging literature on game-theoretic statistics [14, 19]. Aside from the fact that our decision procedure in §5 is based on supermartingales and e-processes, we highlight several connections to related concepts like the e-posterior in §6.

2 Setting

Throughout the paper, denote the per-round outcome set by $\mathcal{Y}$. Unless otherwise stated, we assume $\mathcal{Y}$ is a finite set for expository reasons. For example, we will refer to $X : \mathcal{Y}^T \rightarrow \mathbb{R}$ as a *variable* rather than a random variable, as all such X are measurable. See Example 4.3 and §5 for more general spaces such as $\mathcal{Y} = \mathbb{R}$. Denote the $\mathcal{Y}$ -valued sequences up to time $t - 1$ by $\mathcal{Y}^{<t} = \bigcup_{s=0}^{t-1} \mathcal{Y}^s$ where $\mathcal{Y}^0 = \{\epsilon\}$ is the empty sequence.

2.1 Sets of probabilities, conditioning, and rectangularity

In this paper we imagine the agent’s belief can be expressed as a set of probability measures $\mathcal{P} \subseteq \Delta(\mathcal{Y}^T)$. One central question is how to *update* such a belief upon seeing the value of $Y_{1..t}$. A natural and well-studied way to update $\mathcal{P}$ is to condition each $P \in \mathcal{P}$ on the observation, taking $\mathcal{P}|_{y_{1..t}} := \{P(\cdot \mid Y_{1..t} = y_{1..t})\}_{P \in \mathcal{P}}$. Most of the literature on time consistent robust decision making invokes conditions such as *rectangularity*, under which this conditioning leads to sensible decisions.

DEFINITION 2.1 (RECTANGULARITY). A set of probability measures $\mathcal{P} \subseteq \Delta(\mathcal{Y}^T)$ is rectangular (or stable under pasting [6]) if it is closed under the following operation. For any $t \leq T - 1$, any history $\bar{y} \in \mathcal{Y}^t$, and any $P, Q \in \mathcal{P}$ with $P(\bar{y}) > 0$ and $Q(\bar{y}) > 0$, the measure R defined by

$$R(y_{1..T}) = \begin{cases} P(\bar{y}) \cdot Q(y_{t+1..T} \mid \bar{y}) & \text{if } y_{1..t} = \bar{y}, \\ P(y_{1..T}) & \text{if } y_{1..t} \neq \bar{y} \end{cases}$$

is also in $\mathcal{P}$.

See Bartl [1], Föllmer and Penner [6] for more general definitions for infinite $\mathcal{Y}$. Rectangularity is also closely related to fork convexity [16].

2.2 Upper expectations and other operators

A belief expressed as a set of probability measures $\mathcal{P}$ naturally leads to an operator to assess the badness of a loss random variable $X : \mathcal{Y}^T \rightarrow \mathbb{R}$, the *upper expectation with respect to $\mathcal{P}$* ,

$$\bar{\mathbb{E}}_{\mathcal{P}} X := \sup_{P \in \mathcal{P}} \mathbb{E}_P X, \quad (1)$$

$$\bar{\mathbb{E}}_{\mathcal{P}|Y_{1..t}} X = \sup_{P \in \mathcal{P}} \mathbb{E}_P [X \mid Y_{1..t} = y_{1..t}]. \quad (2)$$

As we will see in the following subsection, such an operator is a useful tool to separate the assessment of a loss random variable from the choice of an action to minimize that measurement. To that end, it is fruitful to consider more general operators and their conditional versions,

$$\bar{\mathbb{E}} : \mathbb{R}^{\mathcal{Y}^T} \rightarrow \mathbb{R}, \quad (3)$$

$$\bar{\mathbb{E}}^c : \mathbb{R}^{\mathcal{Y}^T} \times \mathcal{Y}^t \rightarrow \mathbb{R}, \quad (4)$$

where we write $\bar{\mathbb{E}}^c [X \mid y_{1..t}]$ for the conditional operator. We also sometimes write $\bar{\mathbb{E}}^c [X \mid Y_{1..t} = y_{1..t}]$ for clarity. We define $\bar{\mathbb{E}}^c [X \mid \varepsilon] := \bar{\mathbb{E}} X$ and $\bar{\mathbb{E}}^c [X \mid y_{1..T}] := X(y_{1..T})$. For $X \in \mathbb{R}^{\mathcal{Y}^T}$, we define $\bar{\mathbb{E}} X := \bar{\mathbb{E}} \hat{X}$, where $\hat{X} \in \mathbb{R}^{\mathcal{Y}^T}$ is the extension $\hat{X}(y_{1..T}) = X(y_{1..t})$.

We will consider the following axioms on these general operators. Let $X, X' \in \mathbb{R}^{\mathcal{Y}^T}$.

AXIOM 1 (TRANSLATION OF $\bar{\mathbb{E}}$). $\bar{\mathbb{E}}[X + \alpha] = \bar{\mathbb{E}}X + \alpha$ for all $\alpha \in \mathbb{R}$.

AXIOM 2 (MONOTONICITY OF $\bar{\mathbb{E}}$). $X \geq X' \implies \bar{\mathbb{E}}X \geq \bar{\mathbb{E}}X'$.

These first two axioms are also the defining properties of financial risk measures [8, §4].

AXIOM 3 (MONOTONICITY OF $\bar{\mathbb{E}}^c$). For all t and all $y_{1..t} \in \mathcal{Y}^t$, we have $X(y_{1..t}, \cdot) \geq X'(y_{1..t}, \cdot) \implies \bar{\mathbb{E}}^c [X \mid y_{1..t}] \geq \bar{\mathbb{E}}^c [X' \mid y_{1..t}]$.

Note that Axiom 3 implies that $\bar{\mathbb{E}}^c [X \mid y_{1..t}]$ depends on X only via $X(y_{1..t}, \cdot)$.

To illustrate, let us verify that upper expectations $\bar{\mathbb{E}}_{\mathcal{P}}$ satisfy these three axioms for all $\mathcal{P}$. The first two follow directly from translation and monotonicity of the expectation operator. Axiom 3 similarly follows for all $y_{1..t}$ with $\sup_{P \in \mathcal{P}} P(y_{1..t}) > 0$.

Finally, while not an axiom, the tower property turns out to be intimately related to our study of time consistency. We write $\bar{\mathbb{E}}^c [X \mid Y_{1..t}]$ for the function $\mathcal{Y}^t \ni y_{1..t} \mapsto \bar{\mathbb{E}}^c [X \mid y_{1..t}]$, which as noted above we can extend to all of $\mathcal{Y}^T$.

DEFINITION 2.2 (TOWER PROPERTY). We say $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ satisfy the tower property if for all t and all $X \in \mathbb{R}^{\mathcal{Y}^T}$, we have $\bar{\mathbb{E}}[\bar{\mathbb{E}}^c [X \mid Y_{1..t}]] = \bar{\mathbb{E}}X$.

2.3 Decision problems

We will capture decision problems the agent faces as follows. Let $\mathcal{A}$ be a set of actions. After choosing T actions and observing T outcomes, the agent will suffer a loss given by $L : \mathcal{A}^T \times \mathcal{Y}^T \rightarrow \mathbb{R}$. See Protocol 1.

Protocol 1: Decision protocol

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for  $t = 1, 2, \dots, T$  do
  Agent chooses  $a_t \in \mathcal{A}$ 
  Outcome  $y_t \in \mathcal{Y}$  is revealed
end
Agent suffers loss  $L(\{a_t\}, \{y_t\})$ 

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We can encode the sequential choices of the agent in an algorithm or *decision rule* $\text{Alg} : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$, which chooses $a_t = \text{Alg}(y_{1..t-1})$. Given a decision rule and loss, we write the *loss variable* $L_{\text{Alg}} : \mathcal{Y}^T \rightarrow \mathbb{R}$ given by $L_{\text{Alg}}(y) = L(\{\text{Alg}(y_{1..t-1})\}_{t=1}^T, y)$. That is, as Alg fixes all the contingent choices of the agent, L_{Alg} denotes the resulting loss as a function of the outcome sequence.

A natural choice of loss function simply sums per-round losses $\ell : \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}$, as $L(\{a_t\}, \{y_t\}) = \sum_{t=1}^T \ell(a_t, y_t)$. Two natural examples follow.

- (1) $\mathcal{Y} = \mathcal{A} = \{0, 1\}$, $\ell(a, y) = \mathbb{1}\{a \neq y\}$, $L(\{a_t\}, \{y_t\}) = \sum_t \mathbb{1}\{a_t \neq y_t\}$. See Example 2.7.
- (2) $\mathcal{Y} = \mathcal{A} = \mathbb{R}$, $\ell(a, y) = (a - y)^2$, $L(\{a_t\}, \{y_t\}) = \sum_t (a_t - y_t)^2$. See Example 4.3.

We will often find it useful to simplify to the following *reduced form* setting, which matches the setting of Grünwald and Halpern [12]. Here $T = 2$ and we assume L depends only on a_2 and y_2 , so we may write $L : \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}$. Similarly, the decision rule simplifies to $\text{Alg} : \mathcal{Y} \rightarrow \mathcal{A}$, so that $a_2 = \text{Alg}(y_1)$. The loss random variable is now simply $L_{\text{Alg}}(y) = L(\text{Alg}(y_1), y_2)$. This setting is much simpler yet often captures the essence of the tension between belief updating and time consistency: the agent faces only a single decision, but can formulate a plan before or after observing information.

2.4 Decisions from operators: optimality and time consistency

Let $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ be a generic operator and its conditional version. Let us see how an agent can use these operators to make decisions. In a non-sequential setting, given loss $L : \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}$, the agent can simply choose

$$a^* \in \underset{a \in \mathcal{A}}{\operatorname{argmin}} \bar{\mathbb{E}}[L(a, \cdot)] . \quad (5)$$

That is, the agent can use the operator $\bar{\mathbb{E}}$ to evaluate the badness of the loss following a given action $a \in \mathcal{A}$, and select the least bad action.

This logic applies sequentially as well. Let us illustrate in the reduced form setting, where as above $T = 2$ and the loss $L : \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}$ depends only on a_2 and y_2 . At time $t = 2$ the agent has already received some information in the form of $y_1 \in \mathcal{Y}$. With the conditional version $\bar{\mathbb{E}}^c$, the agent can choose an action conditioned on this information,

$$a_2^*(y_1) \in \underset{a \in \mathcal{A}}{\operatorname{argmin}} \bar{\mathbb{E}}^c[L(a, \cdot) \mid y_1] . \quad (6)$$

Letting $\text{Alg}(y_1) = a^*(y_1)$, these conditional actions together form a decision rule. Similarly, a priori, we can simply choose

$$\text{Alg}^* \in \underset{\text{Alg} : \mathcal{Y} \rightarrow \mathcal{A}}{\operatorname{argmin}} \bar{\mathbb{E}}[L_{\text{Alg}}] . \quad (7)$$

The notion of *time consistency* (Definition 2.5) states that the a posteriori choices encoded in Alg are also a priori optimal.

Let us now define these notions of optimality in our general setting. At time $t = 1$ we define the *a priori optimal* decision rule to be the one that looks best to $\bar{\mathbb{E}}$. We may easily generalize the notion to ϵ -optimality.

DEFINITION 2.3. *Given an operator $\bar{\mathbb{E}}$ and loss L , a decision rule $\text{Alg}' : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$ is ϵ -a priori optimal if*

$$\bar{\mathbb{E}}[L_{\text{Alg}'}] \leq \inf_{\text{Alg} : \mathcal{Y}^{<T} \rightarrow \mathcal{A}} \bar{\mathbb{E}}[L_{\text{Alg}}] + \epsilon. \quad (8)$$

We say Alg' is simply a priori optimal if it is 0-a priori optimal.

Similarly, an *a posteriori optimal* decision rule at time t is a choice of the remaining actions given the previous information $y_{1..t-1}$ and actions chosen $a_{1..t-1}$. Rather than considering conditional decision rules $\text{Alg} : \mathcal{Y}^{<T-t} \rightarrow \mathcal{A}$ that choose only the remaining actions, it will be convenient to keep all decision rules of the form $\mathcal{Y}^{<T} \rightarrow \mathcal{A}$. To that end, we say Alg, Alg' are equivalent up to time t , denoted $\text{Alg} \equiv_t \text{Alg}'$, if $\text{Alg}(y) = \text{Alg}'(y)$ for all $y \in \mathcal{Y}^{<t}$. In other words, $\text{Alg} \equiv_t \text{Alg}'$ if for all $y \in \mathcal{Y}^T$ the sequences $a_s = \text{Alg}(y_{1..s-1})$ and $a'_s = \text{Alg}'(y_{1..s-1})$ are the same for $s = 1, \dots, t$.

DEFINITION 2.4. *Let time $t \leq T$, conditional operator $\bar{\mathbb{E}}^c$, loss L , and reference decision rule $\text{Alg}^* : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$ be given. A decision rule $\text{Alg}_t : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$ is ϵ -a posteriori optimal with respect to Alg^* if $\text{Alg}_t \equiv_{t-1} \text{Alg}^*$ and for all $y_{1..t} \in \mathcal{Y}^t$ we have*

$$\bar{\mathbb{E}}^c[L_{\text{Alg}_t} \mid y_{1..t}] \leq \inf_{\text{Alg} \equiv_{t-1} \text{Alg}^*} \bar{\mathbb{E}}^c[L_{\text{Alg}} \mid y_{1..t}] + \epsilon. \quad (9)$$

We say Alg_t is a posteriori optimal if it is 0-a posteriori optimal.

Note that ϵ -a posteriori optimality at time $t = 1$ is exactly ϵ -a posteriori optimality.

A natural question is whether observing information $y_{1..t}$ and making a fresh decision that minimizes $\bar{\mathbb{E}}^c$ would lead to a decision rule which is a priori optimal. This is exactly the statement of time consistency of the operators.

DEFINITION 2.5. *Operators $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ are time consistent if there exists $c \geq 0$ such that for all $\epsilon, \delta \geq 0$, any ϵ -a priori optimal decision rule Alg^* and time $t \leq T$, any Alg_t which is δ -a posteriori optimal with respect to Alg^* is $(c(\epsilon + \delta))$ -a priori optimal.¹*

In particular, to be time consistent, it must be the case that any a posteriori optimal decision rule following an a priori optimal rule is itself a priori optimal (taking $\epsilon = \delta = 0$). In the reduced form setting, this fully-optimal time consistency simplifies to the condition that if for all $y_1 \in \mathcal{Y}$ we have $\text{Alg}(y_1) \in \argmin_{a \in \mathcal{A}} \bar{\mathbb{E}}^c[L_{\text{Alg}} \mid Y_1]$, then we must have $\text{Alg} \in \argmin_{\text{Alg}' : \mathcal{Y} \rightarrow \mathcal{A}} \bar{\mathbb{E}}[L_{\text{Alg}}]$.

Finally, while not our focus, we will also discuss *dynamic consistency*, which says that the preference order over variables must be consistent.

DEFINITION 2.6. *Operators $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ are dynamically consistent if for all $X, X' : \mathcal{Y}^T \rightarrow \mathbb{R}$ and t we have $\bar{\mathbb{E}}^c[X \mid Y_{1..t}] \leq \bar{\mathbb{E}}^c[X' \mid Y_{1..t}] \implies \bar{\mathbb{E}}X \leq \bar{\mathbb{E}}X'$.*

¹Our definition of time consistency requires the blow-up of the approximation factor to be at most linear, but this condition could be easily relaxed to super-linear factors.

2.5 Limits of simple conditioning

To illustrate our definitions, let us present a running example. This simple Bernoulli setting is also sufficient to see why simple conditioning (updating $\mathcal{P}$ to $\mathcal{P}|_{Y_{1..t}}$) fails to be time consistent.

Example 2.7 (Bernoulli). Consider the reduced form setting with $\mathcal{Y} = \mathcal{A} = \{0, 1\}$, $T = 2$, and 0-1 loss $L(a, y) = \mathbb{1}\{a \neq y\}$. Let $\mathcal{P} = \{\text{Ber}(\epsilon)^2, \text{Ber}(1 - \epsilon)^2\}$ for some $\epsilon \in (0, 1/2)$. (Here and elsewhere, for $P \in \Delta(\mathcal{Y})$, we denote the i.i.d. P distribution by P^t .) For operators, let us consider the upper expectation $\bar{\mathbb{E}}X = \bar{\mathbb{E}}_{\mathcal{P}}X$ and the upper conditional expectation $\bar{\mathbb{E}}^c[X | Y] = \bar{\mathbb{E}}_{\mathcal{P}|_{Y_1}}Y$.

The a priori optimal decision rule minimizes the worst-case expected loss before any observations. In this binary setting, there are only four decision rules to consider: $\text{Alg}^1 : y \mapsto 1$, $\text{Alg}^0 : y \mapsto 0$, $\text{Alg}^{\text{same}} : y \mapsto y$ and $\text{Alg}^{\text{opposite}} : y \mapsto 1 - y$. The upper expected loss of Alg^1 is

$$\bar{\mathbb{E}}[L_{\text{Alg}^1}] = \sup_{P \in \mathcal{P}} \mathbb{E}_P[L_{\text{Alg}^1}] = \max_{P \in \{\text{Ber}(\epsilon)^2, \text{Ber}(1-\epsilon)^2\}} \mathbb{E}_P[\mathbb{1}\{Y_2 = 0\}] = \max(1 - \epsilon, \epsilon) = 1 - \epsilon.$$

By symmetry, $\bar{\mathbb{E}}[L_{\text{Alg}^0}] = 1 - \epsilon$ as well. Similar calculations reveal $\bar{\mathbb{E}}[L_{\text{Alg}^{\text{same}}}] = 2\epsilon(1 - \epsilon)$ and $\bar{\mathbb{E}}[L_{\text{Alg}^{\text{opposite}}}] = \epsilon^2 + (1 - \epsilon)^2$. We conclude that Alg^{same} is a priori optimal.

Let us calculate the a posteriori optimal decision rule. Since the distributions are i.i.d., conditioning yields $\mathcal{P}|_{y_1} = \{\text{Ber}(\epsilon), \text{Ber}(1 - \epsilon)\}$, regardless of y_1 . Thus for any $a \in \mathcal{A}$ we have

$$\begin{aligned} \bar{\mathbb{E}}^c[L(a, Y_2) | y_1] &= \sup_{P \in \mathcal{P}|_{y_1}} \mathbb{E}_P[L(a, Y_2)] \\ &= \sup_{P \in \{\text{Ber}(\epsilon), \text{Ber}(1-\epsilon)\}} \mathbb{E}_P[\mathbb{1}\{a \neq Y_2\}] \\ &= \max(\epsilon, 1 - \epsilon), \end{aligned}$$

which is independent of a . In other words, *every* decision rule is a posteriori optimal in this setting, regardless of the observation. In particular, Alg^1 is a posteriori optimal. As Alg^1 is not a priori optimal, these operators are not time consistent for this $\mathcal{P}$.

3 Tower property is equivalent to time consistency

In this section, we uncover a very close connection between time consistency and the tower property: under our axioms, the two are equivalent. As a consequence, one can take existing results establishing the tower property of upper expectations under rectangularity to recover time consistency in that case. In the following section, we will explore the converse: to show that time consistency is impossible for some $\mathcal{P}$ (when $\bar{\mathbb{E}} = \bar{\mathbb{E}}_{\mathcal{P}}$ is the upper expectation with the respect to $\mathcal{P}$), one can simply show that no conditional operator $\bar{\mathbb{E}}^c$ satisfying the tower property exists.

3.1 Tower property implies time consistency

The intuition that the tower property gives time consistency is straightforward. For any decision rule Alg , we have $\bar{\mathbb{E}}[\bar{\mathbb{E}}^c[L_{\text{Alg}} | Y_{1..t}]] = \bar{\mathbb{E}}[L_{\text{Alg}}]$. Thus if L_{Alg} looks weakly better than all $L_{\text{Alg}'}$ a posteriori, it will also look weakly better a priori.

THEOREM 3.1. *Let $\bar{\mathbb{E}}$ satisfy Axioms 1 (translation) and 2 (monotonicity). If $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ satisfy the tower property, then they are time consistent.*

PROOF. Let decision rule Alg^* be ϵ -a priori optimal, and $t \leq T$. Let $\text{Alg}_t \equiv \text{Alg}^*$.

Alg_t δ -a posteriori optimal

$$\implies \forall y_{1..t} \in \mathcal{Y}^t, \bar{\mathbb{E}}^c[L_{\text{Alg}_t} \mid y_{1..t}] \leq \inf_{\text{Alg} \equiv_{t-1} \text{Alg}^*} \bar{\mathbb{E}}^c[L_{\text{Alg}} \mid y_{1..t}] + \delta$$

$$\implies \forall \text{Alg} \equiv_t \text{Alg}^*, \bar{\mathbb{E}}^c[L_{\text{Alg}_t} \mid Y_{1..t}] \leq \bar{\mathbb{E}}^c[L_{\text{Alg}} \mid Y_{1..t}] + \delta,$$

as variables $\mathcal{Y}^t \rightarrow \mathbb{R}$. Extending to $\mathcal{Y}^T \rightarrow \mathbb{R}$,

$$\implies \forall \text{Alg} \equiv_t \text{Alg}^*, \bar{\mathbb{E}}[\bar{\mathbb{E}}^c[L_{\text{Alg}_t} \mid Y_{1..t}]] \leq \bar{\mathbb{E}}[\bar{\mathbb{E}}^c[L_{\text{Alg}} \mid Y_{1..t}] + \delta] \quad \text{Axiom 2}$$

$$\implies \forall \text{Alg} \equiv_t \text{Alg}^*, \bar{\mathbb{E}}[\bar{\mathbb{E}}^c[L_{\text{Alg}_t} \mid Y_{1..t}]] \leq \bar{\mathbb{E}}[\bar{\mathbb{E}}^c[L_{\text{Alg}} \mid Y_{1..t}]] + \delta \quad \text{Axiom 1}$$

$$\implies \forall \text{Alg} \equiv_t \text{Alg}^*, \bar{\mathbb{E}}[L_{\text{Alg}_t}] \leq \bar{\mathbb{E}}[L_{\text{Alg}}] + \delta \quad \text{tower property}$$

$$\implies \bar{\mathbb{E}}[L_{\text{Alg}_t}] \leq \bar{\mathbb{E}}[L_{\text{Alg}^*}] + \delta \leq \inf_{\text{Alg}: \mathcal{Y}^{<T} \rightarrow \mathcal{A}} \bar{\mathbb{E}}[L_{\text{Alg}}] + \epsilon + \delta \quad \epsilon\text{-a priori optimality}$$

$$\implies \text{Alg}_t \text{ } (\epsilon + \delta)\text{-a priori optimal.}$$

□

The same general logic shows that the tower property also gives the intuitively stronger property of dynamic consistency: if $\bar{\mathbb{E}}[X \mid Y_{1..t}] \leq \bar{\mathbb{E}}[X' \mid Y_{1..t}]$, then $\bar{\mathbb{E}}X \leq \bar{\mathbb{E}}X'$. Theorem 3.1 essentially gives a quantitative version of this statement.

The work of Bartl [1] shows that when $\mathcal{P}$ is rectangular, the upper expectation $\bar{\mathbb{E}}X = \bar{\mathbb{E}}_{\mathcal{P}}X$ and upper conditional expectation $\bar{\mathbb{E}}^c[X \mid Y_{1..t}] = \bar{\mathbb{E}}_{\mathcal{P}|Y_{1..t}}X$ satisfy the tower property. (In fact, the same work establishes the converse as well: these two operators satisfy the tower property exactly when $\mathcal{P}$ is rectangular.) We thus succinctly recover time consistency and dynamic consistency for these operators.

3.2 Time consistency implies tower property

We now show a converse: if $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ are time consistent, then they must satisfy the tower property. In fact, only the exactly optimal version of time consistency ($\epsilon = \delta = 0$) is needed for this result.

The intuition is the following. Suppose the tower rule is violated for some variable X . In particular, suppose X looks strictly better a priori than it does a posteriori. Then one can define a new variable X' that looks even better a posteriori, but because of the tower rule mismatch, still is strictly worse than X a priori. Now define a simple two-action loss, where the action switches between X and X' : the result is that any a posteriori optimal decision rule cannot be a priori optimal. The case where X looks strictly worse a priori follows similarly.

THEOREM 3.2. *Suppose $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ satisfy Axioms 1, 2, and 3. Then time consistency implies the tower property.*

PROOF. Suppose for some X and $t \leq T$ we had $\bar{\mathbb{E}}[\bar{\mathbb{E}}^c[X \mid Y_{1..t}]] \neq \bar{\mathbb{E}}X$. Let us begin with the case $\bar{\mathbb{E}}[\bar{\mathbb{E}}^c[X \mid Y_{1..t}]] > \bar{\mathbb{E}}X$. Define $\hat{X} : \mathcal{Y}^t \rightarrow \mathbb{R}$ by $\hat{X}(y_{1..t}) = \bar{\mathbb{E}}^c[X \mid y_{1..t}]$. Then by the assumption, we have $\bar{\mathbb{E}}\hat{X} > \bar{\mathbb{E}}X$. By Axiom 1 (translation of $\bar{\mathbb{E}}$), there is some $\epsilon > 0$ such that $X' := \hat{X} - \epsilon$ satisfies $\bar{\mathbb{E}}X' = \bar{\mathbb{E}}[\hat{X} - \epsilon] = \bar{\mathbb{E}}\hat{X} - \epsilon > \bar{\mathbb{E}}X$.

Let us now define a decision problem. Let $\mathcal{A} = \{0, 1\}$, and define the loss $L : \mathcal{A}^T \times \mathcal{Y}^T \rightarrow \mathbb{R}$ by $L(a, \cdot) = X$ if $a_{t+1} = 0$ and $L(a, \cdot) = X'$ if $a_{t+1} = 1$. In particular, only at time $t + 1$ does the decision influence the loss. Accordingly, as with the reduced form discussed above, we may write decision rules more simply as $\text{Alg} : \mathcal{Y}^t \rightarrow \mathcal{A}$. Let Alg_0 and Alg_1 be the constant decision rules 0 and 1, i.e., with $\text{Alg}_a(y_{1..t}) = a$ for $a \in \mathcal{A}$ and all $y_{1..t} \in \mathcal{Y}^t$. Note that $L_{\text{Alg}_0} = X$ and $L_{\text{Alg}_1} = X'$ by design.

As $\bar{\mathbb{E}}[L_{\text{Alg}_1}] > \bar{\mathbb{E}}[L_{\text{Alg}_0}]$, it cannot be the case that Alg_1 is a priori optimal. Now let $\text{Alg} : \mathcal{Y}^t \rightarrow \mathcal{A}$ be any a posteriori optimal decision rule. (Just as in the reduced form setting, since the loss depends only on a_{t+1} , prior actions have no effect, and the notion of a posteriori optimality does not require a reference decision rule.) If $\text{Alg} = \text{Alg}_1$ we are done. Otherwise, let $y_{1..t} \in \mathcal{Y}^t$ be any value such that $\text{Alg}(y_{1..t}) = 0$. Then $\bar{\mathbb{E}}^c[L_{\text{Alg}} | y_{1..t}] = \bar{\mathbb{E}}^c[X | y_{1..t}]$ by Axiom 3 (monotonicity of $\bar{\mathbb{E}}^c$), as $L_{\text{Alg}}(y_{1..t}, y_{t+1..T}) = L(0, (y_{1..t}, y_{t+1..T})) = X(y_{1..t}, y_{t+1..T})$. Similarly, we have $\bar{\mathbb{E}}^c[L_{\text{Alg}_1} | y_{1..t}] = \bar{\mathbb{E}}^c[X' | y_{1..t}]$ by Axiom 3. Thus $\bar{\mathbb{E}}^c[L_{\text{Alg}_1} | y_{1..t}] = \bar{\mathbb{E}}^c[X' | y_{1..t}] \leq \bar{\mathbb{E}}^c[X | y_{1..t}] = \bar{\mathbb{E}}^c[L_{\text{Alg}} | y_{1..t}]$. As this inequality holds for all such $y_{1..t} \in \mathcal{Y}^t$, we conclude that L_{Alg_1} is also a posteriori optimal, and $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ fail to be time consistent.

Now suppose $\bar{\mathbb{E}}[\bar{\mathbb{E}}^c[X | Y_{1..t}]] < \bar{\mathbb{E}}X$. Define $\hat{X}$ as above and $X' = \hat{X} + \epsilon$ for $\epsilon > 0$ sufficiently small that $\bar{\mathbb{E}}X' = \bar{\mathbb{E}}[\hat{X} + \epsilon] < \bar{\mathbb{E}}X$. By the same choice of loss, we have $\bar{\mathbb{E}}[L_{\text{Alg}_1}] < \bar{\mathbb{E}}[L_{\text{Alg}_0}]$, so Alg_0 is not a priori optimal. Once again Alg_0 is a posteriori optimal, however, by the same logic as above: for all Alg and all $y_{1..t}$ such that $\text{Alg}(y_{1..t}) = 1$, we have $\bar{\mathbb{E}}^c[L_{\text{Alg}_0} | y_{1..t}] = \bar{\mathbb{E}}^c[X | y_{1..t}] \leq \bar{\mathbb{E}}^c[X' | y_{1..t}] = \bar{\mathbb{E}}^c[L_{\text{Alg}} | y_{1..t}]$. Once again $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ fail to be time consistent. $\square$

As we will see, Theorem 3.2 is a powerful tool to establish the *impossibility* of time consistency: rather than constructing decision problems, one can simply find a violation of the tower property. We will use this perspective extensively in the following section.

4 Impossibility beyond rectangularity

This section analyzes the limitations of proposed updating methods for general sets of probabilities $\mathcal{P}$. We then prove a fundamental impossibility: for large families of non-rectangular $\mathcal{P}$, there is no conditional operator $\bar{\mathbb{E}}^c$ that satisfies the tower property with $\bar{\mathbb{E}} = \bar{\mathbb{E}}_{\mathcal{P}}$ being the upper expectation. In §3 we showed that any such operators must satisfy the tower property for decisions using them to be time consistent. As a result, there is no general way to “update” $\mathcal{P}$, to some object $\mathcal{P}[Y_{1..t}]$ based on observations $Y_{1..t}$, which leads to time-consistent decision making for any downstream decision problem.

4.1 Limits of weighted approaches

Recall from Example 2.7 that simple conditioning, i.e., defining $\bar{\mathbb{E}}^c[X | Y_1] = \bar{\mathbb{E}}_{\mathcal{P}|_{Y_1}} X$, does not lead to time consistency, even for a simple i.i.d. Bernoulli example. In part to remedy the deficiencies of simple conditioning, Halpern and Leung [13] propose an alternative approach: maintain not only the conditional probabilities $P(\cdot | Y_{1..t} = y_{1..t})$ but also weights $\alpha_P(y_{1..t})$ that are updated by the likelihood of the observation. While this likelihood updating method can yield time consistency in some cases, it fails as a general solution.

Example 4.1 (Likelihood weighted Bernoulli). We show likelihood updating yields time consistency for the setting of Example 2.7. Recall that we have the reduced form setting with $\mathcal{Y} = \mathcal{A} = \{0, 1\}$, $T = 2$, 0-1 loss $L(a, y) = \mathbb{1}\{a \neq y\}$, and $\mathcal{P} = \{\text{Ber}(\epsilon)^2, \text{Ber}(1 - \epsilon)^2\}$ for $\epsilon \in (0, 1/2)$. Again we consider the upper expectation $\bar{\mathbb{E}}X = \bar{\mathbb{E}}_{\mathcal{P}}X$, but now the likelihood-weighted upper conditional expectation $\bar{\mathbb{E}}_{\alpha}^c[X | y_1] = \sup_{P \in \mathcal{P}} \alpha_P(y_1) \mathbb{E}_P[X | Y_1 = y_1]$ where $\alpha_P(y_1) = P(Y_1 = y_1)$. The a priori optimal rule remains $\text{Alg}^{\text{same}} : y_1 \mapsto y_1$.

Let us calculate the a posteriori optimal rule Alg_w^* . For $P = \text{Ber}(p)^2$ the likelihood weights are simply $\alpha_{\text{Ber}(p)^2}(1) = p$ and $\alpha_{\text{Ber}(p)^2}(0) = 1 - p$. Since each $P \in \mathcal{P}$ is i.i.d., we have $\mathbb{E}_P[L(a, Y_2) |$

$y_1] = \mathbb{E}_P[L(a, Y_2)] = P(Y_2 = 1 - a)$. Fixing $y_1 = 0$, we now have

$$\begin{aligned}\bar{\mathbb{E}}_\alpha^c[L(0, Y_2) \mid 0] &= \max\{(1 - \epsilon)\epsilon, \epsilon(1 - \epsilon)\} = \epsilon(1 - \epsilon), \\ \bar{\mathbb{E}}_\alpha^c[L(1, Y_2) \mid 0] &= \max\{(1 - \epsilon)^2, \epsilon^2\} = (1 - \epsilon)^2.\end{aligned}$$

We conclude $\text{Alg}_w^*(0) = 0$. By symmetry, $\text{Alg}_w^*(1) = 1$. Thus, $\text{Alg}_w^* = \text{Alg}^{\text{same}}$, and we obtain time consistency.

The following example modifies the previous to effectively duplicate $y = 1$. The idea is to create spurious likelihood spikes that have no relevance to the decision problem at hand, and thus cause the weighted operator to place too much weight on some elements of $\mathcal{P}$. As we will see in Theorem 4.6, the inability for the operator to know the relevant information for the decision problem is inherent.

Example 4.2 (Duplicated Bernoulli). Consider the reduced form setting with $\mathcal{Y} = \{0, 1, 2, 3\}$, $\mathcal{A} = \{0, 1\}$, $T = 2$, and 0-1 loss $L(a, y) = \mathbb{1}\{a \neq [y]\}$ where $[y] = \mathbb{1}\{y \geq 1\}$. Thus 1, 2, 3 are three “copies” of the label 1. Let $\mathcal{P} = \{(P^0)^2, (P^1)^2\} = \{(\frac{2}{3}, \frac{1}{3}, 0, 0)^2, (\frac{1}{3}, \frac{2}{9}, \frac{2}{9}, \frac{2}{9})^2\}$. As the loss effectively ignores the identities of $\{1, 2, 3\}$, one can check that the a priori optimal decision rule is simply $\text{Alg}^{\text{class}} : y \mapsto [y]$, analogous to Alg^{same} .

Now let us calculate the weighted a posteriori optimal rule Alg_w^* . Intuitively, the issue arises when $y_1 = 1$: the weights will set $\alpha_{(P^1)^2}(1) = \frac{2}{9}$ and $\alpha_{(P^0)^2}(1) = \frac{1}{3}$, thereby putting too much weight on the distribution that says $\{1, 2, 3\}$ is unlikely. Concretely, we have

$$\bar{\mathbb{E}}_\alpha^c[L(0, Y_2) \mid 1] = \max(\frac{2}{9} \cdot \frac{2}{3}, \frac{1}{3} \cdot \frac{1}{3}) = \frac{4}{27}, \quad \bar{\mathbb{E}}_\alpha^c[L(1, Y_2) \mid 1] = \max(\frac{2}{9} \cdot \frac{1}{3}, \frac{1}{3} \cdot \frac{2}{3}) = \frac{2}{9}.$$

so $\text{Alg}_w^*(1) = 0$. As $\text{Alg}^{\text{class}}(1) = 1$, the a posteriori optimal rule Alg_w^* is not a priori optimal, and likelihood updating fails time consistency.

Finally, to illustrate the general failure of likelihood updating, consider the following example with $\mathcal{Y} = \mathbb{R}$. Here the a posteriori optimal decision rule disagrees with the a priori optimal rule with probability 1.

Example 4.3 (Normal distributions). Consider the reduced form setting with $\mathcal{Y} = \mathcal{A} = \mathbb{R}$, $T = 2$, $\mathcal{P} = \{\mathcal{N}(-1, 1)^2, \mathcal{N}(1, 1)^2\}$, and squared loss $L(a, y) = (a - y)^2$. We show in Appendix A that the a priori optimal decision rule is $\text{Alg}^*(y_1) = \tanh(y_1)$, while the weighted a posteriori optimal rule is given by a piecewise function, as illustrated in Figure 1b. This demonstrates that likelihood updating fails to preserve time consistency even in continuous settings.

4.2 A general impossibility result

As we have seen, even for simple i.i.d. families $\mathcal{P}$, previously proposed conditional operators all lead to time-inconsistent decision rules for some decision problem. We now generalize this phenomenon to a sweeping impossibility result: in fact there is *no* reasonable conditional operator that yields time consistency for arbitrary $\mathcal{P}$.

As a warm-up, let us sketch the proof that no such operator exists even for the simple Bernoulli setup of Example 2.7.

Example 4.4 (Impossibility for Example 2.7). Again let $\mathcal{Y} = \{0, 1\}$ and $\mathcal{P} = \{\text{Ber}(\epsilon)^2, \text{Ber}(1 - \epsilon)^2\}$ for some $\epsilon \in (0, 1/2)$. Let $\bar{\mathbb{E}} = \bar{\mathbb{E}}_\mathcal{P}$ be the upper expectation, and let $\bar{\mathbb{E}}^c$ be some conditional operator

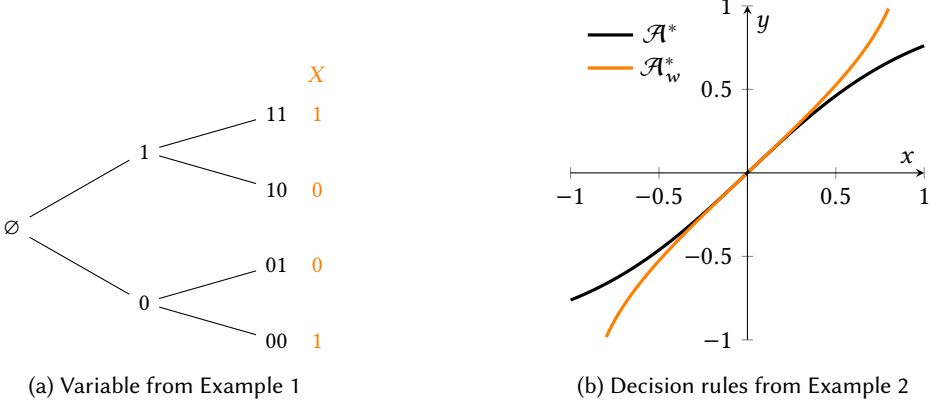


Fig. 1

satisfying Axiom 3 (conditional monotonicity), such that $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ satisfy the tower property. One can show in this example that for any $\alpha \in \mathbb{R}$ we must have $\bar{\mathbb{E}}^c[\alpha \mid y_1] = \alpha$ for all $y_1 \in \mathcal{Y}$.²

Now consider the variable $X(y_1, y_2) = \mathbb{1}\{y_1 = y_2\}$ depicted in Figure 1a. Let $c_y := \bar{\mathbb{E}}^c[X \mid Y_1 = y] = \bar{\mathbb{E}}^c[X \mathbb{1}_{Y_1=y} \mid Y_1 = y]$ by Axiom 3. We have $c_y \geq 0$ by observation $\bar{\mathbb{E}}^c[0 \mid Y_1] = 0$ above and monotonicity. The tower property for $X \mathbb{1}_{Y_1=y}$ gives

$$\begin{aligned} (1 - \epsilon)^2 &= \bar{\mathbb{E}}_{\mathcal{P}} X \mathbb{1}_{Y_1=y} = \bar{\mathbb{E}}_{\mathcal{P}} [\bar{\mathbb{E}}^c[X \mathbb{1}_{Y_1=y} \mid Y_1]] \\ &= \max(\epsilon c_y + (1 - \epsilon)0, (1 - \epsilon)c_y + \epsilon 0) \\ &= (1 - \epsilon)c_y, \end{aligned}$$

giving $c_y = 1 - \epsilon$. The tower property for X now gives

$$\begin{aligned} (1 - \epsilon)^2 + \epsilon^2 &= \bar{\mathbb{E}}_{\mathcal{P}} X = \bar{\mathbb{E}}_{\mathcal{P}} [\bar{\mathbb{E}}^c[X \mid Y_1]] = \max(\epsilon c_0 + (1 - \epsilon)c_1, (1 - \epsilon)c_0 + \epsilon c_1) \\ &= 1 - \epsilon, \end{aligned}$$

giving $\epsilon = 1/2$. As we assumed $\epsilon < 1/2$, we conclude that the tower property is incompatible with Axiom 3 for this example.

We now generalize this impossibility to any i.i.d. family $\mathcal{P}$ with full support. To do so, it is convenient to assume one more axiom that captures reasonable conditional operators: the conditional operator of zero is zero. (This axiom comes for free in Example 4.4.) This axiom is much weaker than other typical axioms, such as assuming $\inf_{y \in \mathcal{Y}^T} X(y) \leq \bar{\mathbb{E}}^c[X \mid Y_{1..t}] \leq \sup_{y \in \mathcal{Y}^T} X(y)$.

AXIOM 4 (CONDITIONAL ZERO). $\bar{\mathbb{E}}^c[0 \mid y_{1..t}] = 0$ for all t and all $y_{1..t} \in \mathcal{Y}^t$.

This axiom allows us to completely pin down $\bar{\mathbb{E}}^c$ in terms of $\mathcal{P}$. The key statement for this result is the identity $\bar{\mathbb{E}}^c[X \mathbb{1}_{y_{1..t}} \mid Y_{1..t}] = \bar{\mathbb{E}}^c[X \mid y_{1..t}] \mathbb{1}_{y_{1..t}}$, since both have zero components for $y'_{1..t} \neq y_{1..t}$.

²Let $c_y(x) = \bar{\mathbb{E}}^c[x \mid Y_1 = y]$ for $x \in \mathbb{R}$ and $y \in \mathcal{Y}$. Let $f(a, b) = \max(\epsilon a + (1 - \epsilon)b, (1 - \epsilon)a + \epsilon b) = \epsilon(a + b) + (1 - 2\epsilon) \max(a, b)$. Then the tower property says $f(c_0(a), c_1(b)) = f(a, b)$ for all $a, b \in \mathbb{R}$. The c_y must be monotone, and in fact strictly increasing and surjective as f is in both arguments. Fixing b and taking $a \rightarrow \infty$ now forces $(1 - \epsilon)c_0(a) + \epsilon c_1(b) = (1 - \epsilon)a + \epsilon b$ for sufficiently large a , regardless of b . Thus $c_1(b) = b + z$ for some $z \in \mathbb{R}$. Taking $b \rightarrow \pm\infty$ gives both $c_0(a) = a - \frac{1-\epsilon}{\epsilon}z$ and $c_0(a) = c_0(a) = a - \frac{\epsilon}{1-\epsilon}z$. As $\epsilon < 1/2$, we must have $z = 0$.

LEMMA 4.5. Let $\mathcal{P} \subseteq \Delta(\mathcal{Y}^T)$ and set $\bar{\mathbb{E}}X = \bar{\mathbb{E}}_{\mathcal{P}}X$. Suppose $\bar{\mathbb{E}}^c$ satisfies Axioms 3 (conditional monotonicity) and 4 (conditional zero), and $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ satisfy the tower property. Then for all $y_{1..t} \in \mathcal{Y}^t$ we have

$$\bar{\mathbb{E}}^c[X \mid y_{1..t}] = \sup_{P \in \mathcal{P}} \frac{P(y_{1..t})}{\mu(y_{1..t})} \mathbb{E}_P[X \mid y_{1..t}], \quad (10)$$

where $\mu(y_{1..t}) = \sup_{P \in \mathcal{P}} P(y_{1..t})$.

PROOF. From Axiom 3, we have $\bar{\mathbb{E}}^c[X \mathbb{1}_{y_{1..t}} \mid y_{1..t}] = \bar{\mathbb{E}}^c[X \mid y_{1..t}]$, and $\bar{\mathbb{E}}^c[X \mathbb{1}_{y_{1..t}} \mid y'_{1..t}] = \bar{\mathbb{E}}^c[0 \mid y'_{1..t}] = 0$ for all $y'_{1..t} \neq y_{1..t}$, the latter equality from Axiom 4. Thus, $\bar{\mathbb{E}}^c[X \mathbb{1}_{y_{1..t}} \mid Y_{1..t}] = \bar{\mathbb{E}}^c[X \mid y_{1..t}] \mathbb{1}_{y_{1..t}}$. From the tower property, we have

$$\begin{aligned} \bar{\mathbb{E}}_{\mathcal{P}}[X \mathbb{1}_{y_{1..t}}] &= \bar{\mathbb{E}}_{\mathcal{P}}[\bar{\mathbb{E}}^c[X \mid y_{1..t}] \mathbb{1}_{y_{1..t}}] \\ &= \bar{\mathbb{E}}^c[X \mid y_{1..t}] \bar{\mathbb{E}}_{\mathcal{P}}[\mathbb{1}_{y_{1..t}}] \\ &= \bar{\mathbb{E}}^c[X \mid y_{1..t}] \mu(y_{1..t}). \end{aligned}$$

By definition, we also have

$$\bar{\mathbb{E}}_{\mathcal{P}}[X \mathbb{1}_{y_{1..t}}] = \sup_{P \in \mathcal{P}} P(y_{1..t}) \mathbb{E}_P[X \mid y_{1..t}].$$

Combining and rearranging terms gives the result. $\square$

We now state our impossibility result. Here P has *full support* if $P(y) > 0$ for all $y \in \mathcal{Y}$.

THEOREM 4.6. Let $\mathcal{P} \subseteq \Delta(\mathcal{Y}^T)$ be a finite set of i.i.d. probability measures with full support, with $T \geq 2$ and $|\mathcal{P}| \geq 2$. Set $\bar{\mathbb{E}}X = \bar{\mathbb{E}}_{\mathcal{P}}X$, and let $\bar{\mathbb{E}}^c$ be any conditional operator satisfying Axioms 3 and 4. Then $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ do not satisfy the tower property.

PROOF. Take $T = 2$ for simplicity (or equivalently, restrict to $\mathcal{Y}^2$). For $y \in \mathcal{Y}$ for brevity we will write $\mu(y) = \mu(Y_1 = y) = \sup_{P \in \mathcal{P}} P(Y_1 = y)$. Let $X = \mathbb{1}\{Y_1 = Y_2\} = \sum_{y \in \mathcal{Y}} \mathbb{1}\{Y_1 = Y_2 = y\}$. From Lemma 4.5 we have $\bar{\mathbb{E}}^c[X \mid Y_1] = \sup_{P \in \mathcal{P}} P(y_1)^2 / \mu(y_1) = \mu(y_1)$. The tower expectation is

$$\bar{\mathbb{E}}_{\mathcal{P}}[\bar{\mathbb{E}}^c[X \mid Y_1]] = \sup_{P \in \mathcal{P}} \sum_{y \in \mathcal{Y}} P(y) \mu(y).$$

We also have

$$\bar{\mathbb{E}}_{\mathcal{P}}X = \sup_{P \in \mathcal{P}} \sum_{y \in \mathcal{Y}} P(y)^2.$$

As $\mathcal{P}$ is finite, let P^* attain the supremum in $\bar{\mathbb{E}}_{\mathcal{P}}X$. We have $P^*(y) \leq \mu(y)$ for all $y \in \mathcal{Y}$ by definition. As $|\mathcal{P}| \geq 2$, we must also have some $y^* \in \mathcal{Y}$ such that $P^*(y^*) < \mu(y^*)$. As P^* has full support, $P^*(y^*) > 0$. Thus

$$\begin{aligned} \bar{\mathbb{E}}_{\mathcal{P}}X &= \sum_{y \in \mathcal{Y}} P^*(y)^2 \\ &< \sum_{y \in \mathcal{Y}} P^*(y) \mu(y) \\ &\leq \sup_{P \in \mathcal{P}} \sum_{y \in \mathcal{Y}} P(y) \mu(y) \\ &= \bar{\mathbb{E}}_{\mathcal{P}}[\bar{\mathbb{E}}^c[X \mid Y_1]]. \end{aligned}$$

$\square$

The conditions of Theorem 4.6 can be weakened. For example, all that is needed is that some P^* achieving the supremum in the L2 norm among $P \in \mathcal{P}$ has full support, or weaker still, that some such P^* has $P^*(y^*) > 0$ for some y^* with $P^*(y^*) < \mu(y^*)$.

Combined with Theorem 3.2, we now have the following.

COROLLARY 4.7. *Let $\mathcal{P}, \bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ satisfy the conditions of Theorem 4.6. Then $\bar{\mathbb{E}}, \bar{\mathbb{E}}^c$ are not time consistent.*

Returning to the discussion in §1.1, Corollary 4.7 implies the impossibility of defining time-consistent dynamic risk measures when the robust representation of the original risk measure (at time $t = 1$) involves sufficiently non-rectangular sets $\mathcal{P}$.

4.3 Implications for updating sets of probabilities

Let us now turn to the question of how to update a belief expressed as a set of probability measures $\mathcal{P} \subseteq \Delta(\mathcal{Y}^T)$. Specifically, we would like to define a conditional version $\mathcal{P}[Y_{1..t}]$, an object that expresses the belief after observing the value of $Y_{1..t}$. So far we have discussed direct conditioning, $\mathcal{P}[Y_{1..t}] = \mathcal{P}|_{Y_{1..t}} := \{P(\cdot | Y_{1..t})\}_{P \in \mathcal{P}}$, and weighted versions $\mathcal{P}[Y_{1..t}] = (\mathcal{P}|_{Y_{1..t}}, \{\alpha_P(Y_{1..t})\}_{P \in \mathcal{P}})$ for some choice of weights $\alpha_P(Y_{1..t})$. For each of these definitions, there are natural choices of $\bar{\mathbb{E}}^c$ to assess the badness of losses, and thus operationalize the belief to make decisions.

Unfortunately, while these objects are natural definitions of $\mathcal{P}[Y_{1..t}]$, and the corresponding conditional operators $\bar{\mathbb{E}}^c$ natural ways to operationalize them, we have seen that they lead to time-inconsistent decisions for some $\mathcal{P}$ and some decision problem. This inconsistency is more than a mathematical inconvenience: it shows that defining $\mathcal{P}[Y_{1..t}]$ and $\bar{\mathbb{E}}^c$ in this way does not lead to a minimal notion of rationality.

These negative results lead to the following general question: does there exist some definition of $\mathcal{P}[Y_{1..t}]$ such that, for some operator $\bar{\mathbb{E}}^c[X | Y_{1..t}]$ which depends only on $\mathcal{P}[Y_{1..t}]$ and $X(Y_{1..t}, \cdot)$, the pair $\bar{\mathbb{E}} = \bar{\mathbb{E}}_{\mathcal{P}}$ and $\bar{\mathbb{E}}^c$ are time consistent for all $\mathcal{P}$ and all decision problems? Perhaps an even more clever weight function $\alpha_P(Y_{1..t})$, or another approach entirely? Theorem 4.6 implies that no such $\mathcal{P}[Y_{1..t}]$ can exist, even when restricting to i.i.d. families $\mathcal{P}$.

5 Time consistency via a robust top-up operator

Borrowing ideas from game-theoretic statistics, we now introduce a decision procedure which optimizes a conditional operator based on the replication cost of the loss random variable. In this approach, the agent maintains an *account* C_t^P for each distribution $P \in \mathcal{P}$, representing a fictitious capital balance. At each time step, the agent chooses a decision rule Alg for the remaining actions, as well as a gambling strategy for each P : using fair gambles under P , and the money in C_t^P , the agent must ensure that each account has enough to cover the loss L_{Alg} by time T . The intuition is that accounts C_t^P with a lot of money relative to the loss correspond to worlds P which are “less likely” relative to the decision at hand. In other words, the inverse of C_t^P behaves as a likelihood weight—but unlike the pitfalls of that approach we saw in Example 4.2, the accounts C_t^P naturally adapt to information relevant to the decision problem.

5.1 Set-up

Due to our reliance on tools from game-theoretic statistics, which easily apply to much more general spaces than finite $\mathcal{Y}$, let us revisit our core definitions. Throughout this section, let $\mathcal{Y}$ and $\mathcal{A}$ be Polish spaces denoting the set of outcomes and actions respectively. Fix a time horizon $T \in \mathbb{N}$. We take $\Omega \subseteq \mathcal{Y}^T$ with the canonical process $Y_t(\omega) = \omega_t$ and filtration $\mathcal{F}_t = \sigma(Y_{1..t})$. Note: $\mathcal{F}_0 = \{\emptyset, \Omega\}$ is the trivial sigma algebra. Let $\mathcal{P}$ denote a set of probability measures. As is common

in game-theoretic statistics and robust probability [3, 11, 16, 19], we assume the existence of a dominating measure Q for $\mathcal{P}$, i.e., for any $N \in \mathcal{F}_T$, $Q(N) = 0 \implies P(N) = 0, \forall P \in \mathcal{P}$.

The interaction proceeds for $t = 1, 2, \dots, T$ as follows: at the start of round t the agent chooses an action a_t as a (measurable) function of the past outcomes $Y_{1..t-1}$, then the outcome Y_t is revealed. After the final round, the agent suffers a loss

$$L(a_{1..T}, y_{1..T}) \in L^1(\mathcal{F}_T),$$

where $L : \mathcal{A}^T \times \mathcal{Y}^T \rightarrow \mathbb{R}$.

As before, a decision rule is a map $\text{Alg} : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$, which specifies a_t given the history $y_{1..t-1}$, so that $a_t = \text{Alg}(y_{1..t-1})$. The induced terminal loss random variable is

$$L_{\text{Alg}} : \Omega \rightarrow \mathbb{R}, \quad L_{\text{Alg}}(y_{1..T}) := L\left(\{\text{Alg}(y_{1..t-1})\}_{t=1}^T, y_{1..T}\right) \in L^1(\mathcal{F}_T).$$

Thus, the total loss incurred by Alg is the terminal random variable L_{Alg} .

5.2 Supermartingales and the Initial Capital Operator

We recall the notion of supermartingales, based on which we introduce the *initial capital operator* which relates to the *initial capital measure* defined in Ruf et al. [17].

Supermartingales are stochastic processes whose expected value conditional on present information, can only weakly decrease over time. A natural interpretation is that they model fair-or-unfavorable bets.

DEFINITION 5.1 (P- SUPERMARTINGALE). *A process $M = (M_t)_{t \in \mathbb{N}_0}$ adapted to some filtration $(\mathcal{F}_t)_{t \in \mathbb{N}_0}$ is a P-supermartingale; i.e., for all $0 \leq s \leq t$,*

$$\mathbb{E}_P[M_t \mid \mathcal{F}_s] \leq M_s \quad P\text{-a.s.}$$

We now define the initial capital operator, which will be vital for constructing the time-consistent decision procedure developed in subsequent sections. Informally, it is a robust operator: it gives the minimal initial value needed so that a suitable family of supermartingales can replicate a given random quantity of interest.

DEFINITION 5.2 (INITIAL CAPITAL OPERATOR). *Fix a time horizon T and let $X \in \mathcal{F}_T$. Define the Inverse Capital Operator of X as follows:*

$$v(X) := \text{ess inf}_Q \left\{ \alpha \in \mathcal{F}_0 : \forall P \in \mathcal{P} \exists (A_u^P)_{u=0}^t \text{ a } P\text{-supermartingale, } A_0^P \leq \alpha, A_T^P \geq X \right\}.$$

When X is an indicator random variable, our definition differs slightly from the initial-capital measure of Ruf et al. [17], which is formulated in terms of e-processes rather than supermartingales. (Specifically, we restrict to e-processes of the form $E_t = \inf_P M_t^P$, which do not generally cover all e-processes [15].) Nonetheless, Proposition 5.3 shows that our initial-capital operator coincides with the upper expectation; the same equivalence holds for the initial-capital measure as well in our finite-time setting [17].

Following Ruf et al. [17], we will use the notation $\mu(X) := \bar{\mathbb{E}}_{\mathcal{P}} X$ to denote the upper expectation of X with respect to $\mathcal{P}$. The following result shows that $v(X) = \mu(X)$ in our setting. Thus, taking $\bar{\mathbb{E}}X = v(X)$ in the sequel still represents the usual upper expectation.

PROPOSITION 5.3 (INITIAL CAPITAL OPERATOR EQUALS UPPER EXPECTATION). *Assume $\mathcal{F}_0 = \{\emptyset, \Omega\}$ is trivial and $X \in L^1(P)$ for every $P \in \mathcal{P}$. Then*

$$v(X) = \mu(X).$$

PROOF. Since $\mathcal{F}_0$ is trivial, any $\alpha \in \mathcal{F}_0$ is a constant, so the ess inf_Q in the definition of $v(X)$ coincides with the usual infimum over constants.

We first show that $v(X) \leq \mu(X)$. Fix $P \in \mathcal{P}$ and define the Doob martingale

$$A_t^P := \mathbb{E}_P[X \mid \mathcal{F}_t], \quad t = 0, 1, \dots, T.$$

Then $(A_t^P)_{t=0}^T$ is a P -martingale, hence a P -supermartingale, and

$$A_T^P = X, \quad A_0^P = \mathbb{E}_P[X \mid \mathcal{F}_0] = \mathbb{E}_P[X],$$

where the last equality uses that $\mathcal{F}_0$ is trivial. Thus for any constant $\alpha \geq \mu(X) = \sup_{P \in \mathcal{P}} \mathbb{E}_P[X]$ we have, for every $P \in \mathcal{P}$, a P -supermartingale (A_t^P) with $A_0^P = \mathbb{E}_P[X] \leq \alpha$ and $A_T^P = X \geq X$. Hence α is feasible in the set defining $v(X)$, and taking the infimum over such α yields $v(X) \leq \mu(X)$.

Now, we prove the other direction, i.e., $v(X) \geq \mu(X)$. Let α be any feasible constant in the set defining $v(X)$. By feasibility, for each $P \in \mathcal{P}$ there exists a P -supermartingale $(A_t^P)_{t=0}^T$ such that $A_0^P \leq \alpha$ and $A_T^P \geq X$. Since (A_t^P) is a P -supermartingale,

$$A_0^P \geq \mathbb{E}_P[A_T^P] \geq \mathbb{E}_P[X].$$

Combining with $A_0^P \leq \alpha$ gives $\alpha \geq \mathbb{E}_P[X]$ for every $P \in \mathcal{P}$, hence $\alpha \geq \sup_{P \in \mathcal{P}} \mathbb{E}_P[X] = \mu(X)$. Since this holds for every feasible α , taking the infimum over feasible α yields $v(X) \geq \mu(X)$.

Putting these together we conclude $v(X) = \mu(X)$. $\square$

5.3 Robust top-up operator

We now define our conditional operator, $\Delta_t(X; C_t)$. Unlike $\bar{\mathbb{E}}^c$, the operator Δ_t depends on “accounts” C_t^P for each $P \in \mathcal{P}$. If the current balances C_t are too small to guarantee replication of a terminal target X , we must inject extra capital. We introduce the *robust top-up operator*, which formalizes the smallest such injection needed to ensure some supermartingale family can reach X by time T .

Let $t \leq T$. In what follows, $C_t^P \in \mathcal{F}_t$ will be used to denote the realized accounts at time t for each $P \in \mathcal{P}$. Compactly, we will use $C_t := \{C_t^P\}_{P \in \mathcal{P}}$ to denote the state of accounts at time t . We will find it useful to consider the set of supermartingale families that start from C_t and replicate X by time T .

DEFINITION 5.4 (SUPERMARTINGALE FAMILIES FOR $C_t \rightarrow X$). Let $t \leq T$, let C_t denote accounts realized at time t , and suppose $X \in \mathcal{F}_T$. Define $S_t(X; C_t)$ to be the collection of families $\{(A_u^P)_{u=t}^T\}_{P \in \mathcal{P}}$ such that for every $P \in \mathcal{P}$:

- (1) $(A_u^P)_{u=t}^T$ is a P -supermartingale;
- (2) $A_t^P \leq C_t^P$
- (3) $A_T^P \geq X$.

The set $S_t(X; C_t)$ comprises of all supermartingale families that can replicate X at time T , while requiring a simultaneous top-up of at most δ to the accounts at time t . It is possible that for some $P \in \mathcal{P}$, C_t^P does not suffice to replicated X by time T . We therefore define the minimal additional amount needed to replicate X as the smallest amount α for which $S_t(X; C_t + \alpha)$ is nonempty. We formalize this below via the *robust top-up operator*.

DEFINITION 5.5 (ROBUST TOP-UP OPERATOR). Let $t \leq T$, let C_t denote the accounts realized at time t , and let $X \in \mathcal{F}_T$. Define

$$\Delta_t(X; C_t) := \text{ess inf}_Q \{ \alpha \in \mathcal{F}_t : S_t(X; C_t + \alpha) \neq \emptyset \}.$$

In particular, when $t = 0$ and $C_0 = \mathbf{0}$, we define $v(X) := \Delta_0(X; \mathbf{0})$.

Note: We may sometimes use $\Delta_{s,s'}$ for arbitrary $s \leq s'$ to generalize this notion to timelines beginning at s and ending at s' .

We show that (v, Δ) satisfy monotonicity and the tower-property, which in turn facilitate a decision procedure that satisfies time consistency, as we shall see in the subsequent subsection. We refer the reader to appendix B for statements and proofs of these properties.

5.4 A time-consistent decision procedure

We now define optimality and time consistency for our operators. For the conditional operator Δ_t , the agent updates a family of accounts over $\mathcal{P}$, and the relevant notion of “value” is the robust initial capital needed to cover the eventual loss. Because exact minimizers need not exist, we work throughout with ϵ -optimality notions both *a priori* and *a posteriori*. We define these below.

DEFINITION 5.6 (ϵ -A PRIORI OPTIMAL). Let $\epsilon \geq 0$. We say an algorithm $\text{Alg}^* : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$ is ϵ -a priori optimal if it satisfies the following:

$$v(L_{\text{Alg}^*}) \leq \text{ess inf}_Q \{v(L_{\text{Alg}}) \mid \text{Alg is a decision rule}\} + \epsilon$$

DEFINITION 5.7 (ϵ -A POSTERIORI OPTIMAL). Let $\epsilon \geq 0$ and let $t \in \{1, \dots, T\}$. Let $\text{Alg}_{t-1} : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$ be a reference decision rule and let $C_{t-1} = \{C_{t-1}^P\}_{P \in \mathcal{P}}$ be an $\mathcal{F}_{t-1}$ -measurable account state at time $t - 1$. We say that a decision rule $\text{Alg}_t : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$ is ϵ -a posteriori optimal at time t (relative to $(\text{Alg}_{t-1}, C_{t-1})$) if:

- (1) $\text{Alg}_t \equiv_{t-1} \text{Alg}_{t-1}$ (i.e., Alg_t makes the same decisions as Alg_{t-1} up to time $t - 1$), and
- (2)

$$\Delta_{t-1,T}(L_{\text{Alg}_t}; C_{t-1}) \leq \text{ess inf}_Q \left\{ \Delta_{t-1,T}(L_{\text{Alg}}; C_{t-1}) : \text{Alg} \equiv_{t-1} \text{Alg}_{t-1} \right\} + \epsilon.$$

With these notions in place, we can describe our time-consistent decision procedure. At each time t , we (i) choose a continuation rule that is nearly optimal given the current account state, and (ii) top up the accounts just enough to ensure the resulting terminal loss can be replicated. The account update is then obtained by evaluating the chosen supermartingale family at the newly observed outcome.

Protocol 2: Decision Procedure

Initialize: set the account state $C_0 = \mathbf{0}$ and choose an arbitrary reference decision rule $\text{Alg}_0 : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$.

for $t = 1, 2, \dots, T$ **do**

Pick an ϵ -a posteriori optimal decision rule $\text{Alg}_t : \mathcal{Y}^{<T} \rightarrow \mathcal{A}$ relative to $(\text{Alg}_{t-1}, C_{t-1})$, i.e.,

$$\Delta_{t-1}(L_{\text{Alg}_t}; C_{t-1}) \leq \text{ess inf}_Q \left\{ \Delta_{t-1}(L_{\text{Alg}}; C_{t-1}) : \text{Alg} \equiv_{t-1} \text{Alg}_{t-1} \right\} + \epsilon.$$

Set $\beta = (\Delta_{t-1}(L_{\text{Alg}_t}; C_{t-1}) + \epsilon)$ and **pick** a β -supermartingale family for $C_{t-1} + \beta \rightarrow X$:

$$\left\{ (A_u^P)_{u=t-1}^T \right\}_{P \in \mathcal{P}} \in S_{t-1}(L_{\text{Alg}_t}; C_{t-1} + \beta).$$

Play the action $a_t := \text{Alg}_t(y_{1:t-1})$.

Observe the outcome $y_t \in \mathcal{Y}$.

Update the account state for the next round:

$$C_t := (A_t^P)_{P \in \mathcal{P}}.$$

end

Agent suffers terminal loss $L_{\text{Alg}_T}(y_{1:T})$.

Protocol 2 satisfies time consistency as described in the theorem below. We provide a proof-sketch of the theorem statement, deferring the complete proof to appendix B.

THEOREM 5.8 (TIME CONSISTENCY OF THE DECISION PROCEDURE). Fix $\epsilon \geq 0$ and a terminal horizon T . Let $(\text{Alg}_t, C_t)_{t=0}^T$ be the sequence generated by the online decision procedure in Protocol 2 (with

$C_0 = \mathbf{0}$). Then the terminal decision rule Alg_T is $(2T - 1)\epsilon$ -a priori optimal, that is:

$$v(L_{\text{Alg}_T}) \leq \text{ess inf}_Q \{v(L_{\text{Alg}}) : \text{Alg} : \mathcal{Y}^{<T} \rightarrow \mathcal{A}\} + (2T - 1)\epsilon.$$

PROOF SKETCH. Let $V^\star := \text{ess inf}_Q \{v(L_{\text{Alg}}) : \text{Alg}\}$.

Let $k \in \{1, \dots, T\}$. For the step k , the protocol chooses Alg_k to be ϵ -a posteriori optimal relative to $(\text{Alg}_{k-1}, C_{k-1})$. Plugging in the feasible competitor Alg_{k-1} gives

$$\Delta_{k-1}(L_{\text{Alg}_k}; C_{k-1}) \leq \Delta_{k-1}(L_{\text{Alg}_{k-1}}; C_{k-1}) + \epsilon \leq \epsilon,$$

where the last inequality holds because C_{k-1} was produced from a supermartingale family that replicates $L_{\text{Alg}_{k-1}}$. Extending the supermartingales backward, and applying the tower property and monotonicity properties, we get that $v(L_{\text{Alg}_t}) \leq \Delta_{0,t-1}(C_{t-1}; 0) + \epsilon$. We then show inductively that

$$\Delta_{0,t-1}(C_{t-1}; 0) \leq V^\star + 2(t - 1)\epsilon, \quad t = 1, \dots, T,$$

where $V^\star := \text{ess inf}_Q \{v(L_{\text{Alg}}) : \text{Alg} : \mathcal{Y}^{<T} \rightarrow \mathcal{A}\}$. Iterating yields $\Delta_{0,T-1}(C_{T-1}; 0) \leq V^\star + 2(T - 1)\epsilon$, and combining with $v(L_{\text{Alg}_T}) \leq \Delta_{0,T-1}(C_{T-1}; 0) + \epsilon$ gives

$$v(L_{\text{Alg}_T}) \leq V^\star + (2T - 1)\epsilon.$$

□

5.5 Examples

We demonstrate how the time-consistent decision procedure operates through two examples. We begin with the two-coin prediction model, where the account updates and continuation choice can be computed explicitly. We then turn to a qualitatively different Gaussian example, illustrating how the same procedure behaves in a continuous outcome space with squared loss.

Example 5.9 (Bernoulli: supermartingale accounts and time-consistent continuation). Let us again continue Examples 2.7, 4.1, 4.4. We are again in the reduced form with $\mathcal{A} = \mathcal{Y} = \{0, 1\}$ (tails/heads), 0–1 loss $L(a, y) = \mathbb{1}\{a \neq y\}$, and

$$\mathcal{P} = \{\text{Ber}(1/3)^2, \text{Ber}(2/3)^2\}.$$

The a priori optimal decision rule is $\text{Alg}^{\text{same}} : y_1 \mapsto y_1$, with terminal loss $L_{\text{Alg}}(y_1, y_2) = \mathbb{1}\{y_1 \neq y_2\}$.

We illustrate the account-based approach: maintain an account for each $P \in \mathcal{P}$, and at each time step choose actions so that all accounts can replicate the relevant terminal loss with as little additional cash as possible.

Write $M_t^{1/3}$ for the capital process under $P = \text{Ber}(1/3)^2$ and $M_t^{2/3}$ for the capital process under $P = \text{Ber}(2/3)^2$. A natural way to replicate L_{Alg} is to take each account to be the conditional expectation of L_{Alg} under its corresponding distribution. For $P = \text{Ber}(2/3)^2$, this yields

$$M_0^{2/3} = \frac{4}{9}, \quad M_1^{2/3}(1) = \frac{1}{3}, \quad M_1^{2/3}(0) = \frac{2}{3},$$

and at time 2,

$$M_2^{2/3}(1, 1) = 0, \quad M_2^{2/3}(1, 0) = 1, \quad M_2^{2/3}(0, 1) = 1, \quad M_2^{2/3}(0, 0) = 0.$$

Similarly, for $P = \text{Ber}(1/3)^2$,

$$M_0^{1/3} = \frac{4}{9}, \quad M_1^{1/3}(1) = \frac{2}{3}, \quad M_1^{1/3}(0) = \frac{1}{3},$$

and

$$M_2^{1/3}(1, 1) = 0, \quad M_2^{1/3}(1, 0) = 1, \quad M_2^{1/3}(0, 1) = 1, \quad M_2^{1/3}(0, 0) = 0.$$

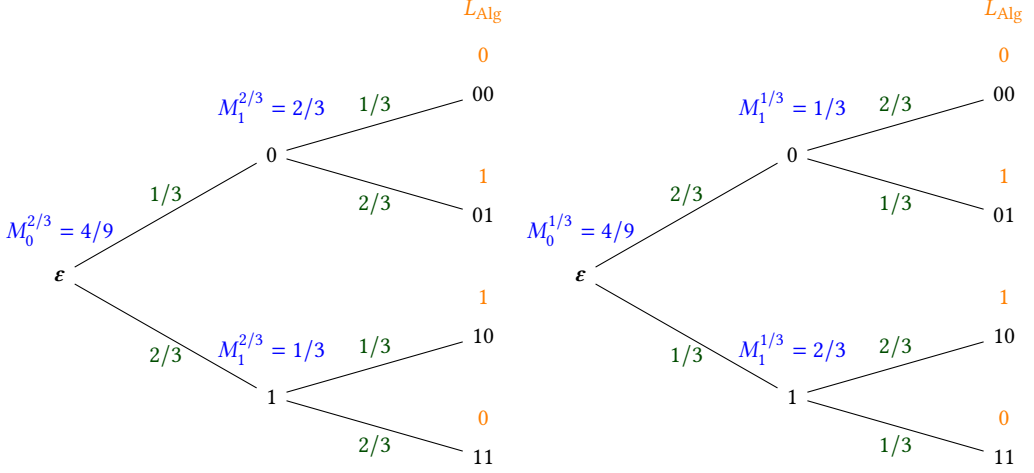


Fig. 2. Supermartingales for the Bernoulli example.

One can check that each $(M_t^P)_{t=0}^2$ is a martingale (hence a supermartingale) under its corresponding $P = \text{Ber}(p)^2$, and that both replicate L_{Alg} at time 2.

Now suppose we observe $y_1 = 1$ (heads). Then $M_1^{2/3}(1) = 1/3$ while $M_1^{1/3}(1) = 2/3$. If we play action $a = 1$, the induced terminal loss is $L(1, y_2) = \mathbb{1}\{y_2 = 0\}$, which can be replicated by both accounts without any further top-up. In contrast, if we play $a = 0$, then $L(0, y_2) = \mathbb{1}\{y_2 = 1\}$, whose conditional expectation given $y_1 = 1$ equals $2/3$ under $\text{Ber}(2/3)^2$ but only $1/3$ under $\text{Ber}(1/3)^2$; ensuring *both* accounts can cover the realized loss requires an additional top-up of $1/3$ in the worst case. Thus the account-based continuation criterion selects $a = 1$ after observing $y_1 = 1$.

The same logic extends to the “duplicated” variant of Example 4.2 with $\mathcal{Y} = \{0, 1, 2, 3\}$. Here instead of betting on individual outcomes, the optimal betting strategy is to follow the $\{0, 1\}$ supermartingales, and only ever bet on the full equivalence classes $[0] = \{0\}$ or $[1] = \{1, 2, 3\}$. Concretely, we take $\hat{M}_1^P(y) = M_1^P([y])$ and similarly for other time steps. As we can see, the optimal betting strategy naturally recovers the correct “weights” on each $P \in \mathcal{P}$, as relevant to the actual decision problem at hand.

Example 5.10. Let us return briefly to Example 4.3, the reduced form setting where $\mathcal{Y} = \mathbb{R}$, $\mathcal{P} = \{N(1, 1)^2, N(-1, 1)^2\}$, and $L(a, y) = (a - y)^2$. Let $P^+ = N(1, 1)^2$ and $P^- = N(-1, 1)^2$. The a priori optimal decision rule is $\text{Alg}^*(y_1) = \tanh y_1$.

The agent initializes $C_0^{P^+} = C_0^{P^-} = 1 + \mathbb{E}_{N(1,1)}(\tanh Y - 1)^2$, and commits to the martingales that correspond to Alg^* :

$$M_1^{P^+}(y_1) = 1 + (\tanh y_1 - 1)^2 \quad (11)$$

$$M_1^{P^-}(y_1) = 1 + (\tanh y_1 + 1)^2 \quad (12)$$

$$M_n^{P^+}(y_1, y_2) = M_n^{P^-}(y_1, y_2) = (\tanh y_1 - y_2)^2. \quad (13)$$

Upon seeing y_1 , the agent updates

$$C_1^{P^+}(y_1) = 1 + (\tanh y_1 - 1)^2 \quad (14)$$

$$C_1^{P^-}(y_1) = 1 + (\tanh y_1 + 1)^2, \quad (15)$$

and chooses $a \in \mathbb{R}$ and $M_2^{P+}(y_2), M_2^{P-}(y_2)$ to minimize

$$\sup_{y_2 \in \mathbb{R}} (a - y_2)^2 - \min \left(\hat{M}_2^{P+}(y_2), \hat{M}_2^{P-}(y_2) \right), \quad (16)$$

subject of course to the supermartingale conditions

$$\mathbb{E}_{P+} \left[\hat{M}_2^{P+}(Y_2) \right] \leq \hat{M}_1^{P+} = C_1^{P+}(y_1) = 1 + (\tanh y_1 - 1)^2, \quad (17)$$

$$\mathbb{E}_{P-} \left[\hat{M}_2^{P-}(Y_2) \right] \leq \hat{M}_1^{P-} = C_1^{P-}(y_1) = 1 + (\tanh y_1 + 1)^2. \quad (18)$$

This objective (16) is minimized by taking $a = \tanh y_1$, $\hat{M}_2^{P+} = M_2^{P+}$, and $\hat{M}_2^{P-} = M_2^{P-}$, giving time consistency.

5.6 Remarks

It is interesting to compare our procedure to the results of §3 and §4. Even though ν and Δ satisfy a tower property, unlike the operators $\bar{\mathbb{E}}^c$ from §3, we do not recover dynamic consistency. While we will have $\Delta_t(X; C_t) \leq \Delta_t(X'; C_t) \implies \nu(X) \leq \nu(X')$ when the accounts C_t are optimal for replicating X , this implication need not hold for arbitrary C_t . As for the impossibility in §4 (Theorem 4.6), that applied to operators that depended only on $X \mid Y_{1..t}$, whereas C_t may not. Similarly, the “updated” $\mathcal{P}$ now includes C_t , which may depend on X .

6 Discussion

We have presented an equivalence between time consistency and the tower property, which led to a general impossibility: there is no way to condition a set of probability measures $\mathcal{P}$ on new information, such that an upper-expectation-like operator leads to time-consistent decisions. To circumvent this impossibility, we presented a novel procedure based on game-theoretic statistics, which maintains accounts to guide future decisions. This latter connection leads to several future directions.

Connection to the e-posterior. Our decision procedure in §5 is closely related to the “e-posterior” minimax decision rule δ^{MM} based of Grünwald [9, 10]. There, just as in our case, a family of (nonnegative) P -supermartingales are chosen for each $P \in \mathcal{P}$, and the inverse value of those supermartingales can be thought of as a sort of likelihood weight for P . Unlike our procedure, however, in these works the decision rule δ^{MM} is derived from the supermartingales rather than the other way around. Furthermore, the assessment of loss a priori is not fixed to be the upper expectation with respect to $\mathcal{P}$. Nonetheless, it would be interesting to understand the time-consistency of δ^{MM} , perhaps using the tools developed here, and generally understand the connection between the two approaches.

Conditional game-theoretic expectations. Game-theoretic statistics is based on *game-theoretic probability*, an alternative branch of probability theory built on game theory rather than measure theory [7, 18, 19]. This theory is built on local gambles, which depend only on the sequence of outcomes seen thus far. For such sequential gamble spaces, the tower property for the game-theoretic conditional (upper) expectation operator always holds [7]. Moreover, when this theory agrees with measure-theoretic probability, it largely does so by representing rectangular sets of probability measures $\mathcal{P}$. An open question has been whether a conditional version of the game-theoretic operator can be defined for *any* gamble space, even without sequential structure. Intuitively, our impossibility result suggests that no such operator could satisfy the tower property.

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A Example 4.3

Consider the reduced form setting $\mathcal{Y} = \mathcal{A} = \mathbb{R}$, $T = 2$, $\mathcal{P} = \{\mathcal{N}(-1, 1)^2, \mathcal{N}(1, 1)^2\}$, and squared loss $L(a, y) = (a - y)^2$. We begin by calculating the *a priori decision rule*:

$$\text{Alg}^* \in \operatorname{argmin}_{\text{Alg}: \mathcal{Y} \rightarrow \mathcal{A}} \sup_{\mu \in \{-1, 1\}} \mathbb{E}_{P_\mu}[(\text{Alg}(Y_1) - Y_2)^2],$$

where under P_μ we have $Y_1, Y_2 \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, 1)$.

Since Y_1 and Y_2 are independent conditional on μ , we can rewrite

$$\mathbb{E}_{P_\mu}[(\text{Alg}(Y_1) - Y_2)^2] = \mathbb{E}_{P_\mu}[(\text{Alg}(Y_1) - \mu)^2] + 1.$$

Expanding and using $\mathbb{E}_{P_\mu}[Y_2] = \mu$ and $\text{Var}_{P_\mu}(Y_2) = 1$ yields

$$\mathbb{E}_{P_\mu}[(\text{Alg}(Y_1) - Y_2)^2] = \mathbb{E}_{P_\mu}[\text{Alg}(Y_1)^2] - 2\mu \mathbb{E}_{P_\mu}[\text{Alg}(Y_1)] + (\mu^2 + 1) = \mathbb{E}_{P_\mu}[(\text{Alg}(Y_1) - \mu)^2] + 1.$$

Thus, the problem reduces to minimax estimation of μ from $Y_1 \sim \mathcal{N}(\mu, 1)$ with $\mu \in \{-1, 1\}$:

$$\min_{\text{Alg}: \mathcal{Y} \rightarrow \mathcal{A}} \max_{\mu \in \{-1, 1\}} \mathbb{E}_{Y_1 \sim \mathcal{N}(\mu, 1)}[(\text{Alg}(Y_1) - \mu)^2].$$

By symmetry, the least favorable prior is uniform on $\{-1, 1\}$, and the minimax estimator equals the Bayes estimator under squared loss, i.e. the posterior mean:

$$\text{Alg}^*(y_1) = \mathbb{E}[\mu \mid Y_1 = y_1].$$

Let $f_\mu(y_1)$ denote the normal density:

$$f_\mu(y_1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y_1 - \mu)^2\right).$$

Then Bayes' rule gives

$$P(\mu = 1 \mid Y_1 = y_1) = \frac{f_1(y_1)}{f_1(y_1) + f_{-1}(y_1)} = \frac{1}{1 + e^{-2y_1}} = \frac{e^{2y_1}}{1 + e^{2y_1}},$$

and $P(\mu = -1 \mid Y_1 = y_1) = \frac{1}{1 + e^{2y_1}}$. Therefore,

$$\text{Alg}^*(y_1) = P(\mu = 1 \mid Y_1 = y_1) - P(\mu = -1 \mid X = y_1) = \frac{e^{2y_1} - 1}{e^{2y_1} + 1} = \tanh(y_1).$$

Now, we turn to the *weighted a posteriori optimal rule*:

$$\text{Alg}_w^*(y_1) \in \operatorname{argmin}_{a \in \mathcal{A}} \sup_{\mu \in \{-1, 1\}} f_\mu(y_1) \cdot \mathbb{E}_{P_\mu}[(a - Y_2)^2 \mid Y_1 = y_1].$$

Since Y_1 is independent of Y_2 given μ and $Y \sim \mathcal{N}(\mu, 1)$, we have

$$\mathbb{E}_{P_\mu}[(a - Y_1)^2 \mid Y_2 = y_2] = \mathbb{E}_{P_\mu}[(a - Y_2)^2] = 1 + (a - \mu)^2.$$

We ignore the common factor $(2\pi)^{-1/2}$ in $f_\mu(y_1)$, and obtain

$$\max \left\{ e^{-\frac{1}{2}(y_1-1)^2} (1 + (a-1)^2), e^{-\frac{1}{2}(y_1+1)^2} (1 + (a+1)^2) \right\}.$$

For $y_1 > 0$, the minimizer occurs when the two terms are equal:

$$e^{-\frac{1}{2}(y_1-1)^2} (1 + (a-1)^2) = e^{-\frac{1}{2}(y_1+1)^2} (1 + (a+1)^2).$$

Expanding and using $\frac{e^{-\frac{1}{2}(y_1-1)^2}}{e^{-\frac{1}{2}(y_1+1)^2}} = e^{2x}$ yields

$$e^{2y_1} (a^2 - 2a + 2) = a^2 + 2a + 2,$$

which implies

$$a^2 - 2 \coth(y_1)a + 2 = 0.$$

Thus for $y_1 \neq 0$,

$$a = \coth(y_1) \pm \sqrt{\coth^2(y_1) - 2}.$$

For $y_1 > 0$, the relevant root is the smaller one,

$$\text{Alg}_w^*(y_1) = \coth(y_1) - \sqrt{\coth^2(y_1) - 2}.$$

Since we restrict actions to $a \in [-1, 1]$, the minimizer cannot exceed ± 1 . For $y_1 > 0$ we obtain the threshold

$$1 = \coth(y_1) - \sqrt{\coth^2(y_1) - 2}, \iff \coth(y_1) = \frac{3}{2} \iff y_1 = \frac{1}{2} \ln 5.$$

By symmetry, the negative threshold is $y_1 = -\frac{1}{2} \ln 5$. Therefore,

$$\text{Alg}_w^*(y_1) = \begin{cases} 1, & y_1 > \frac{1}{2} \ln 5 \\ \coth(y_1) - \sqrt{\coth^2(y_1) - 2}, & 0 < y_1 \leq \frac{1}{2} \ln 5, \\ 0, & y_1 = 0, \\ \coth(y_1) + \sqrt{\coth^2(y_1) - 2}, & -\frac{1}{2} \ln 5 \leq y_1 < 0, \\ -1, & y_1 < -\frac{1}{2} \ln 5. \end{cases}$$

B Time consistent decision procedure via the robust top-up operator

For every $P \in \mathcal{P}$, we define $\rho_{t,T}^P(X)$ as the amount of capital required at time t to replicate X via P -supermartingale updates. This object, while distinct from our robust operator v , will serve as a technical tool to facilitate a tower-rule for v .

DEFINITION B.1. Fix $t \leq T$ and $P \in \mathcal{P}$. For $X \in L^0(\mathcal{F}_T)$ define,

$$\rho_{t,T}^P(X) := \text{ess inf}_P \left\{ m \in L^0(\mathcal{F}_t) : \exists (M_u)_{u=t}^T \text{ a } P\text{-supermartingale with } M_t \leq m, M_T \geq X \right\}.$$

Write the $\mathcal{P}$ -envelope $\rho_{t,T}(X) := (\rho_{t,T}^P(X))_{P \in \mathcal{P}}$.

As before, we consider a scenario where accounts are maintained for each $P \in \mathcal{P}$. For $X \in \mathcal{F}_T$, we will introduce our key operator $v(X)$ to be the (simultaneous) initial capital needed across all $\mathcal{P}$ accounts in order to replicate X by time T via supermartingale updates.

Let $s \leq t \in [T]$. In what follows, $C_s^P \in L_0(\mathcal{F}_s)$ will be used to denote the realized accounts at time s for each $P \in \mathcal{P}$. Compactly, we will use $C_s := \{C_s^P\}_{P \in \mathcal{P}}$ to denote the state of accounts at time s . Let $D_t^P \in L_0(\mathcal{F}_t)$ be a target state, i.e., a state we would like our P -account to realize by time t . Denote by $D_t := \{D_t^P\}_{P \in \mathcal{P}}$ the set of target states at time t . We now define our robust top-up operator v and its conditional version Δ .

DEFINITION B.2. Let $s \leq t \in [T]$ and let C_s and D_t denote realized and target states respectively. Define the minimal additional amount at time s required to guarantee D_t at time t by

$$\Delta_{s,t}(D_t; C_s) := \text{ess inf}_Q \left\{ \alpha \in L^0(\mathcal{F}_s) : \forall P \in \mathcal{P} \exists (A_u^P)_{u=s}^t \text{ a } P\text{-supermartingale, } A_s^P \leq C_s^P + \alpha, A_t^P \geq D_t^P \right\}.$$

For scalar $X \in L^0(\mathcal{F}_t)$ we will sometimes write $\Delta_{s,t}(X; C_s)$ as shorthand for $\Delta_{s,t}(D_t; C_s)$ with $D_t^P \equiv X$ for all P . Furthermore, for $X \in L^0(\mathcal{F}_T)$, we define

$$v(X) := \Delta_{0,T}(X; 0).$$

The key properties that will facilitate time-consistency of ν are its monotonicity and the tower property. We state and prove these below.

LEMMA B.3 (MONOTONICITY). *Let $s \leq t \in [T]$ and let C_s and D_t denote realized and target states respectively. Let $X, X' \in L^0(\mathcal{F}_T)$. Then the following hold:*

- (a) *If $X \leq X'$ P -a.s. then $\rho_{t,T}^P(X) \leq \rho_{t,T}^P(X')$ P -a.s.*
- (b) *If $D_t \leq D'_t$ coordinatewise ($D_t^P \leq D'^P_t$ P -a.s. for each P), then $\Delta_{s,t}(D_t; C_s) \leq \Delta_{s,t}(D'_t; C_s)$ Q -a.s.*
- (c) *If $C_s \leq C'_s$ coordinatewise then $\Delta_{s,t}(D_t; C'_s) \leq \Delta_{s,t}(D_t; C_s)$ Q -a.s.*

PROOF. (a) Fix $P \in \mathcal{P}$ and suppose $X \leq X'$ P -a.s. Let $m \in L^0(\mathcal{F}_t)$ be feasible for $\rho_{t,T}^P(X')$, i.e., there exists a P -supermartingale $(M_u)_{u=t}^T$ such that $M_t \leq m$ and $M_T \geq X'$. Since $X \leq X'$ P -a.s., we also have $M_T \geq X$ P -a.s., so the same m is feasible for $\rho_{t,T}^P(X)$. Thus the feasible set for $\rho_{t,T}^P(X')$ is contained in the feasible set for $\rho_{t,T}^P(X)$, and taking P -essential infima yields $\rho_{t,T}^P(X) \leq \rho_{t,T}^P(X')$ P -a.s.

(b) Suppose $D_t \leq D'_t$ coordinatewise, i.e. for each $P \in \mathcal{P}$, $D_t^P \leq D'^P_t$ P -a.s. Let $\alpha \in L^0(\mathcal{F}_s)$ be feasible for $\Delta_{s,t}(D'_t; C_s)$. By definition, for every $P \in \mathcal{P}$ there exists a P -supermartingale $(A_u^P)_{u=s}^t$ such that

$$A_s^P \leq C_s^P + \alpha, \quad A_t^P \geq D'^P_t.$$

Since $D_t^P \geq D'^P_t$ P -a.s., we also have $A_t^P \geq D_t^P$ P -a.s. Hence the same α is feasible for $\Delta_{s,t}(D_t; C_s)$. Therefore the feasible set defining $\Delta_{s,t}(D'_t; C_s)$ is contained in the feasible set defining $\Delta_{s,t}(D_t; C_s)$, and taking the Q -essential infimum yields

$$\Delta_{s,t}(D_t; C_s) \leq \Delta_{s,t}(D'_t; C_s) \quad (Q\text{-a.s.}).$$

(c) Suppose $C_s \leq C'_s$ coordinatewise, i.e. for each $P \in \mathcal{P}$, $C_s^P \leq C'^P_s$ P -a.s. Let $\alpha \in L^0(\mathcal{F}_s)$ be feasible for $\Delta_{s,t}(D_t; C_s)$. Then for every $P \in \mathcal{P}$ there exists a P -supermartingale $(A_u^P)_{u=s}^t$ such that

$$A_s^P \leq C_s^P + \alpha, \quad A_t^P \geq D_t^P.$$

Since $C_s^P \leq C'^P_s$ P -a.s., we also have $A_s^P \leq C'^P_s + \alpha$. Thus the same α is feasible for $\Delta_{s,t}(D_t; C'_s)$. Therefore the feasible set for $\Delta_{s,t}(D_t; C_s)$ is contained in the feasible set for $\Delta_{s,t}(D_t; C'_s)$, and taking Q -essential infima gives

$$\Delta_{s,t}(D_t; C'_s) \leq \Delta_{s,t}(D_t; C_s) \quad (Q\text{-a.s.}).$$

□

LEMMA B.4 (TOWER PROPERTY). *Fix $s \leq t \in [T]$, $X \in L^0(\mathcal{F}_T)$, and a state C_s at time s . Then*

$$\Delta_{s,T}(X; C_s) = \Delta_{s,t}(\rho_{t,T}(X); C_s).$$

In particular, if $s = 0$ and $C_s = \mathbf{0}_\mathcal{P}$, we have that:

$$\nu(X) = \Delta_{0,t}(\rho_{t,T}(X); \mathbf{0}_\mathcal{P}).$$

PROOF. We prove both inequalities.

($\geq$). Let $\alpha \in L^0(\mathcal{F}_s)$ be feasible for $\Delta_{s,T}(X; C_s)$, i.e., for every $P \in \mathcal{P}$ there exists a P -supermartingale $(M_u^P)_{u=s}^T$ such that

$$M_s^P \leq C_s^P + \alpha \quad \text{and} \quad M_T^P \geq X.$$

Fix $P \in \mathcal{P}$ and consider the tail process $(M_u^P)_{u=t}^T$. This is still a P -supermartingale and it super-replicates X at time T . By definition of $\rho_{t,T}^P(X)$ as the minimal (essential) starting budget at time t needed to super-replicate X by time T via P -supermartingale updates, we must have

$$M_t^P \geq \rho_{t,T}^P(X).$$

Hence the restricted family $(M_u^P)_{u=s}^t$ witnesses that α is feasible for $\Delta_{s,t}(\rho_{t,T}(X); C_s)$ (coordinatewise in P). Therefore

$$\Delta_{s,t}(\rho_{t,T}(X); C_s) \leq \alpha.$$

Taking the Q -essential infimum over all such feasible α yields

$$\Delta_{s,t}(\rho_{t,T}(X); C_s) \leq \Delta_{s,T}(X; C_s).$$

($\leq$). Conversely, suppose $\alpha \in L^0(\mathcal{F}_s)$ is feasible for $\Delta_{s,t}(\rho_{t,T}(X); C_s)$. Then for every $P \in \mathcal{P}$ there exists a P -supermartingale $(A_u^P)_{u=s}^t$ such that

$$A_s^P \leq C_s^P + \alpha \quad \text{and} \quad A_t^P \geq \rho_{t,T}^P(X).$$

Fix $\varepsilon > 0$. By definition of $\rho_{t,T}^P(X)$, there exists a P -supermartingale $(B_u^{P,\varepsilon})_{u=t}^T$ such that

$$B_t^{P,\varepsilon} \leq \rho_{t,T}^P(X) + \varepsilon \quad \text{and} \quad B_T^{P,\varepsilon} \geq X.$$

Now paste (A^P) and $(B^{P,\varepsilon})$ by defining a process $(C_u^{P,\varepsilon})_{u=s}^T$ via

$$C_u^{P,\varepsilon} := \begin{cases} A_u^P, & u \leq t, \\ B_u^{P,\varepsilon} + (A_t^P - B_t^{P,\varepsilon}), & u \geq t. \end{cases}$$

For $u < t$, this is a P -supermartingale because it equals A^P ; for $u \geq t$, it is a P -supermartingale because it equals $B^{P,\varepsilon}$ plus an $\mathcal{F}_t$ -measurable constant shift. Moreover,

$$C_s^{P,\varepsilon} = A_s^P \leq C_s^P + \alpha,$$

and at time T ,

$$C_T^{P,\varepsilon} = B_T^{P,\varepsilon} + (A_t^P - B_t^{P,\varepsilon}) \geq X + (\rho_{t,T}^P(X) - (\rho_{t,T}^P(X) + \varepsilon)) = X - \varepsilon.$$

Finally, define $\tilde{C}_u^{P,\varepsilon} := C_u^{P,\varepsilon} + \varepsilon$ for $u = s, \dots, T$. Then $(\tilde{C}_u^{P,\varepsilon})_{u=s}^T$ is still a P -supermartingale, and it satisfies

$$\tilde{C}_s^{P,\varepsilon} \leq C_s^P + \alpha + \varepsilon, \quad \tilde{C}_T^{P,\varepsilon} \geq X.$$

Thus $\alpha + \varepsilon$ is feasible for $\Delta_{s,T}(X; C_s)$. Since $\varepsilon > 0$ was arbitrary, we get

$$\Delta_{s,T}(X; C_s) \leq \alpha.$$

Taking the Q -essential infimum over feasible α for $\Delta_{s,t}(\rho_{t,T}(X); C_s)$ yields

$$\Delta_{s,T}(X; C_s) \leq \Delta_{s,t}(\rho_{t,T}(X); C_s).$$

Combining the two inequalities gives the desired equality. $\square$

We now define ε -approximate supermartingale families starting at C_t and replicating X by time T .

DEFINITION B.5 (ε -APPROXIMATE SUPERMARTINGALES). Let $\varepsilon \geq 0$ and $X \in L^0(\mathcal{F}_T)$. We say $\{(A_u^P)_{u=t}^T\}_{P \in \mathcal{P}} \in S_t^\varepsilon(X; C_t)$, if for every $P \in \mathcal{P}$, it holds that:

- (1) A^P is a P -supermartingale.
- (2) $A_t^P \leq C_t^P + \Delta_{t,T}(X; C_t) + \varepsilon$.
- (3) $A_T^P \geq X$.

Finally, we state and prove our time consistency theorem.

THEOREM B.6 (TIME CONSISTENCY OF THE DECISION PROCEDURE). Fix $\epsilon \geq 0$ and a terminal horizon T . Let $(\text{Alg}_t, C_t)_{t=0}^T$ be the sequence generated by the online decision procedure in Protocol 2 (with $C_0 = \mathbf{0}$). Then the terminal decision rule Alg_T is $(2T - 1)\epsilon$ -a priori optimal, that is:

$$v(L_{\text{Alg}_T}) \leq \underset{Q}{\text{ess inf}} \{v(L_{\text{Alg}}) : \text{Alg} : \mathcal{Y}^{<T} \rightarrow \mathcal{A}\} + (2T - 1)\epsilon.$$

PROOF. Define the optimal a-priori benchmark

$$V^\star := \underset{Q}{\text{ess inf}} \{v(L_{\text{Alg}}) : \text{Alg} : \mathcal{Y}^{<T} \rightarrow \mathcal{A}\}.$$

Fix a round $t \in \{2, \dots, T\}$. By the protocol, $\text{Alg}_t \equiv_{t-1} \text{Alg}_{t-1}$ and

$$\Delta_{t-1,T}(L_{\text{Alg}_t}; C_{t-1}) \leq \underset{Q}{\text{ess inf}} \left\{ \Delta_{t-1,T}(L_{\text{Alg}}; C_{t-1}) : \text{Alg} \equiv_{t-1} \text{Alg}_{t-1} \right\} + \epsilon.$$

Plugging in the feasible choice $\text{Alg} = \text{Alg}_{t-1}$ (it trivially satisfies $\text{Alg} \equiv_{t-1} \text{Alg}_{t-1}$) yields

$$\Delta_{t-1,T}(L_{\text{Alg}_t}; C_{t-1}) \leq \Delta_{t-1,T}(L_{\text{Alg}_{t-1}}; C_{t-1}) + \epsilon. \quad (19)$$

Now use the fact that, from the previous round ($t - 1$), the protocol picked a supermartingale family

$$\left\{ (A_{t-1,u}^P)_{u=t-2}^T \right\}_{P \in \mathcal{P}} \in S_{t-2}^\epsilon(L_{\text{Alg}_{t-1}}; C_{t-2}),$$

and then updated $C_{t-1} := (A_{t-1,t-1}^P)_{P \in \mathcal{P}}$. In particular, for each $P \in \mathcal{P}$, the tail $(A_{t-1,u}^P)_{u=t-1}^T$ is a P -supermartingale that starts at $A_{t-1,t-1}^P = C_{t-1}^P$ and ends at $A_{t-1,T}^P \geq L_{\text{Alg}_{t-1}}$. Thus it witnesses feasibility of top-up 0 for the conditional problem, so

$$\Delta_{t-1,T}(L_{\text{Alg}_{t-1}}; C_{t-1}) \leq 0. \quad (20)$$

Combining (19) and (20) gives

$$\Delta_{t-1,T}(L_{\text{Alg}_t}; C_{t-1}) \leq \epsilon, \quad t = 2, \dots, T. \quad (21)$$

Next, observe that by definition of $\Delta_{t-1,T}(L_{\text{Alg}_t}; C_{t-1})$, the inequality $\Delta_{t-1,T}(L_{\text{Alg}_t}; C_{t-1}) \leq \epsilon$ means that for each $P \in \mathcal{P}$ there exists a P -supermartingale $(B_u^P)_{u=t-1}^T$ such that

$$B_{t-1}^P \leq C_{t-1}^P + \epsilon + \frac{1}{n}, \quad B_T^P \geq L_{\text{Alg}_t}.$$

But $\rho_{t-1,T}^P(L_{\text{Alg}_t})$ is the P -essential infimum of such starting budgets, hence we conclude

$$\rho_{t-1,T}^P(L_{\text{Alg}_t}) \leq C_{t-1}^P + \epsilon + \frac{1}{n} \implies \rho_{t-1,T}^P(L_{\text{Alg}_t}) \leq C_{t-1}^P + \epsilon \quad \text{for all } P \in \mathcal{P}. \quad (22)$$

Equivalently, as vectors, $\rho_{t-1,T}(L_{\text{Alg}_t}) \leq C_{t-1} + \epsilon \mathbf{e}$ coordinatewise.

Now, by the tower property,

$$v(L_{\text{Alg}_t}) = \Delta_{0,T}(L_{\text{Alg}_t}; \mathbf{0}) = \Delta_{0,t-1}(\rho_{t-1,T}(L_{\text{Alg}_t}); \mathbf{0}).$$

Using monotonicity in the target and (22),

$$v(L_{\text{Alg}_t}) \leq \Delta_{0,t-1}(C_{t-1} + \epsilon \mathbf{e}; \mathbf{0}). \quad (23)$$

Finally, shifting the target vector by a constant ϵ can be paid by adding ϵ at time 0: indeed, adding ϵ to every P -account is itself achieved by the constant (hence supermartingale) process equal to ϵ . Thus

$$\Delta_{0,t-1}(C_{t-1} + \epsilon \mathbf{e}; \mathbf{0}) \leq \Delta_{0,t-1}(C_{t-1}; \mathbf{0}) + \epsilon. \quad (24)$$

Combining (23) and (24) yields

$$v(L_{\text{Alg}_t}) \leq \Delta_{0,t-1}(C_{t-1}; \mathbf{0}) + \epsilon. \quad (25)$$

At this point, we need to bound $\Delta_{0,t-1}(C_{t-1}; 0)$.

Fix $k \in \{1, \dots, t-1\}$. From the protocol, we have a family

$$\left\{ (A_{k,u}^P)^T_{u=k-1} \right\}_{P \in \mathcal{P}} \in S_{k-1}^\epsilon(L_{\text{Alg}_k}; C_{k-1}).$$

By definition of S_{k-1}^ϵ , for each P the process $(A_{k,u}^P)^T_{u=k-1}$ is a P -supermartingale with

$$A_{k,k-1}^P \leq C_{k-1}^P + \Delta_{k-1,T}(L_{\text{Alg}_k}; C_{k-1}) + \epsilon, \quad A_{k,k}^P = C_k^P.$$

Taking conditional expectations under P and using the supermartingale property gives

$$\mathbb{E}_P[C_k^P \mid \mathcal{F}_{k-1}] = \mathbb{E}_P[A_{k,k}^P \mid \mathcal{F}_{k-1}] \leq A_{k,k-1}^P \leq C_{k-1}^P + \beta_k,$$

where we write

$$\beta_k := \Delta_{k-1,T}(L_{\text{Alg}_k}; C_{k-1}) + \epsilon \in L^0(\mathcal{F}_{k-1}). \quad (26)$$

Now define, for each $P \in \mathcal{P}$, the reserve process

$$R_{t-1}^P := 0, \quad R_{k-1}^P := \mathbb{E}_P \left[\sum_{j=k}^{t-1} \beta_j \mid \mathcal{F}_{k-1} \right] \quad \text{for } k = 1, \dots, t-1.$$

(Equivalently, $R_{k-1}^P = \beta_k + \mathbb{E}_P[R_k^P \mid \mathcal{F}_{k-1}]$.) Set

$$M_k^P := C_k^P + R_k^P, \quad k = 0, 1, \dots, t-1.$$

Then for each $k = 1, \dots, t-1$,

$$\begin{aligned} \mathbb{E}_P[M_k^P \mid \mathcal{F}_{k-1}] &= \mathbb{E}_P[C_k^P \mid \mathcal{F}_{k-1}] + \mathbb{E}_P[R_k^P \mid \mathcal{F}_{k-1}] \\ &\leq (C_{k-1}^P + \beta_k) + (R_{k-1}^P - \beta_k) = C_{k-1}^P + R_{k-1}^P = M_{k-1}^P. \end{aligned}$$

Hence $(M_k^P)_{k=0}^{t-1}$ is a P -supermartingale, and by construction $M_{t-1}^P = C_{t-1}^P$ (since $R_{t-1}^P = 0$). Therefore the initial value $M_0^P = R_0^P$ is a feasible top-up at time 0 (in the P -account) to reach C_{t-1}^P by time $t-1$ via the supermartingale $(M_k^P)_{k=0}^{t-1}$. Since the same argument holds for every $P \in \mathcal{P}$, we obtain the robust bound

$$\Delta_{0,t-1}(C_{t-1}; 0) \leq \text{ess sup}_Q \left\{ R_0^P : P \in \mathcal{P} \right\} = \text{ess sup}_Q \left\{ \mathbb{E}_P \left[\sum_{j=1}^{t-1} \beta_j \mid \mathcal{F}_0 \right] : P \in \mathcal{P} \right\}. \quad (27)$$

By definition (26), for $k = 1$ we have $\beta_1 = \Delta_{0,T}(L_{\text{Alg}_1}; 0) + \epsilon = \nu(L_{\text{Alg}_1}) + \epsilon \leq V^\star + 2\epsilon$. For each $k \geq 2$, (21) gives $\Delta_{k-1,T}(L_{\text{Alg}_k}; C_{k-1}) \leq \epsilon$, hence $\beta_k \leq 2\epsilon$.

Therefore, for every $P \in \mathcal{P}$,

$$\mathbb{E}_P \left[\sum_{j=1}^{T-1} \beta_j \mid \mathcal{F}_0 \right] \leq (V^\star + 2\epsilon) + \sum_{j=2}^{T-1} 2\epsilon = V^\star + (2T-2)\epsilon,$$

and so (27) (with $t = T$) yields

$$\Delta_{0,T-1}(C_{T-1}; 0) \leq V^\star + (2T-2)\epsilon.$$

Finally apply (25) with $t = T$:

$$\nu(L_{\text{Alg}_T}) \leq \Delta_{0,T-1}(C_{T-1}; 0) + \epsilon \leq V^\star + (2T-1)\epsilon,$$

which is the desired a-priori guarantee. $\square$